



Forensic Medicine Workbook



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INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.



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Chapter 01 :-

FMT

PM → autopsy = necropsy [necro → dead body] Italy - Bartolomeo

1st → _____

→ Bukley → Wheeler → Arsenic → Chennai

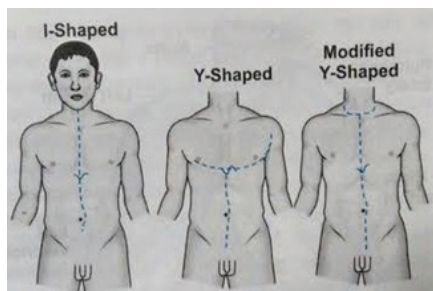
Types of incision:-

M.C → 1. I shape

2. Y shape → Why → ? → FM → Acromian - Process

3. Modified Y → Asphyxial Death = Hanging (Mastoid Process)

4. Inverted Y → Infant



5. X - Shape → Back → Custodial - Death

Organ	Dissection
Stomach	Greater - Curvature
Spinal cord	Posterior
Heart	RA → RV → LA → LV

AUTOPSY (TYPE)

MC →

1. Medicolegal autopsy → Police|Mag. Un-Natural Death=(Whole Body)

2. Pathological autopsy → Relative → Consent → (N) = Cavity

3. Psychological → Suicide → Question = Relative

4. Virtual autopsy → COD → CT/MRI

5. Negative autopsy → PM → Gross → ⊖

No → COD Histo-Path → ⊖

Toxicology → ⊖

2-5% are negative autopsy

If patient dies due to :

A. Vagal inhibition

B. Laryngospasm

6. Obscure autopsy → Not-Sure → COD

PM → Minimal-finding

PM techniques

1. Virchow's method → Very-Very-Common (MC), one by one organ removal

2. Ghons method → (G.B) EN-Block [Cervical, Thoracic, Abdominal, Pelvic]

3. Lettule method → EN-Mass=B.V. Intact [Cervical + Thoracic + Abdominal + Pelvic]

4. Rokitansky method → _____

a) In-SITU

b) In-Side Dissection

Infections = Death

MIST

	1st cavity	Last cavity
Poison	Brain	_____
< 1 year, infant, new Born	Brain-Abd-Thorax	_____
Asphyxial death →	Cranium	Neck = Blood-less
Head injury	Head	_____
Air embolism	Thorax	_____
Pulmonary Air embolism	Thorax	Water test Pyrogallol-test

Under taker # = PM# = _____

→ "After Death"

→ "C₆ - C₇#"

→ Transport → DB (Dead - Body)



FORENSIC PSYCHIATRY

Chapter 02 :-

FORENSIC PSYCHIATRY

Insane → _____

Crime → _____

Not responsible(NR) [unsound mind]

1. 1843 → Mcnaughten-Rule → insane → murder Legal-Test

Insane = mental health care act

= 3 Check-Ups

= (10D) = Mag.

= 30D = Max.

= _____

2. American Rule

3. Brawner Rule

4. Currens Rule = Capacity

5. Durham Rule = Product test

INSANE - SANE - INSANE

Lucid interval → Will

Crime → responsible

Witness → responsible

Sound mind, ≥18 y

WILL → Mental capacity = Testamentary Capacity

Own Handwriting → Holographic-will

3y ← home ← juvenile <18y

Minimum age of criminal responsibility → 7y

Minimum age of full criminal responsibility → 18y

MIST



DEFINITIONS

Chapter 03 :-

DEFINITIONS:-

Impotency → male → Penis Erection, Maintain → Psychogenic,
→ Vasculogenic
→ Neurogenic

Quoad → male → Impotent = Particular - Woman

Frigidity → _____ = FM → Sexual Arousal, Maintain

Divorce → Frigidity > Impotence after marriage, Infertility → Not

Post-humous child = child born after death of father

Suppositious child → Fictitious Child

Fm → Preg → X

Black mail → y = Money

→ superfetation = $2T$ → $2mc$ -cycle =

Twins

→ superfecundation = T → Same mc cycle

2 OVA Fertilized → 2- Coitus
Age → Gap → 1 Month

Same Different

2 OVA Fertilized → 2- Coitus

Same-Age

Same Different

ART (artificial reproduction technique)

Artificial insemination

Donor = semen

= 1ml

Max = 10-Children

surrogacy = law

Type = Altruistic, Medical -

Expenses, Insurance = 3y

Mother → 25-35y

Relative

Indian

One-Child

Once (3-Attempt)

Couple → INDIAN

→ 5y

→ infertility

Consent = Husband + Wife (Donor, Recipient)

Birth → Child = adopt = legal

Hymen = Fold of Mucous-membrane, Mc type = Semilunar 70%

Rupture = Sex, Masturbation, Trauma, Sx, Infection

Postero-lateral

Antero-lateral

=
=
=

1. Young girl → call girl = Sola - PITH = Vagina = Dilate

→ Aptae - VIRIS H = Rupture

2. Girl child → RAPE = hymen not ruptured

Deep

- Glastier - Keen - ROD → Hymen - Examination

- Fimbriated Hymen → look like → Hymen Tear



MIST

Virginity → lost → De Floration

True virgin → No - Sex ($H_{\pm}$)

False virgin → Sex + = H = Intact = Elastic

Parrot

Parous

Nulliparous



Uterine cavity → (Round)

Triangular



External OS → (Transverse)

Circular



Chapter 04 :-

- **MTP** **Up-to** **SETH**
- Contraceptive Fail - 20W → Social
 - Rape, minor, Incest - 24 W → Humanatarian
 - Child Malform - 24 W, > 24W → Eugenic, Teratogenic
 - Medical Board = OBG
 - = Peds
 - = Radio/Sono
 - = Gazzeted officer
 - Mother life = Danger = Any time - 1 Dr. = Therapeutic

Medical Abortion

Indicated legally

Consent = Mother, ≥ 18y

MTP act = 1971

= 2020-21

Done upto

- 24w
- up to 20w → 1Dr.
- 20-24w → 2Dr.

Dr. →

→ RMP

→ 25 → Assist → MTP, 5 Independent

&

Criminal Abortion

Killing a foetus

criminal abortion=DR=family members

→ 6m = House job → obg

→ OBG → MD, MS, DNB, Diploma

Infanticide - Killing a child < 1y

Neonaticide - killing a child within → 24hrs. of birth

Filicide - killing a child by Parents

Viability = fetus

Capable of survival outside → 28-Weeks = 7M

Dead born

Still birth

Live birth

IUD R → Robert sign = Earliest → 12hrs, Gas → Heart, Aorta

Rigor Mortis +

Aseptic autolysis → aka = Maceration

Skin = peeled off

= Blister present on the skin

= ODOR → Sweet

= M. Imp Sign



Mummification → A. fluid ↓, Air ↓, Blood ↓, Odorless (Dry-Hot)

↓
Shrink



Maceration → Spalding sign = Skull - bone - Overlap

Radiological sign → Robert - Sign, Spalding Sign

Ball - Sign = Hyperflexion of spine

LIVE BORN → R (+)

R- {Lung Inflated [Diaphragm → 6-7th ribs]}

a - Lung weight → 30 gm - _____ R+ 60 gm = 2 Fold = Fodere-Test

b - Lung/body weight 1/70 - _____ R+ 1/35 = Proportion = Ploquet test

c - Air in ear → Wreden Test

d - Breslau 2nd Life test → Stomach/Int. ← Air

e - Breslau 1st Life test → Lung ← Air

aka ↓ Floating+ve

Residual volume +

Raygat Test

S. Gravity = 0.94

Hydrostatic Test

Water = specific gravity = 1

Sinking = -ve

BIRTH

ARTHREE

Umblical day (closes)

└─┬─┐	Artery	3D →	medial umbilical fold
	Vein	4D →	ligamentum teres
	Cord	5-6D →	fall

Battered Baby Syndrome → Campe - Syndrome

Caffey syndrome → child = Beaten

→ History → Discrepancy

→ Injury

M. characteristic → Tear - inside - Lips

MC = COD = Head - Injury

Pinch = _____ Butterfly - Bruise

Ribs# = (Nobbing#), String of Bead App.

Skull# = Linear#, Egg-Shell#

Shaking baby syndrome → SDH

aka ↓

→ Intra ocular bleeding

Infantile whiplash syndrome

↓ aka

DOT - BLOT Hem.

SIDS = sudden infant death syndrome/ COT - (Death), CRIB-Death

COD = Sleep-Apnea, Negative autopsy

Card. Arrhythmia

TRACES OF EVIDENCES (CRIME SCENE)

Chapter 05 :-

TRACES OF EVIDENCES (crime scene)

1. Blood stains has a enzyme = Peroxidase

↓
Screening test

old ← M.sensitive ← 1. Bezidine → Blue

Kastle-Myer ← 2. Phenolphthalin → Pink

3. Ortho - Toluidine → Blue - Green

4. LEU Comalachite → Peacock - Blue

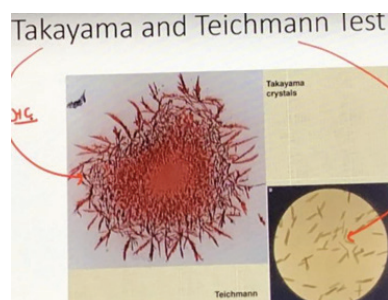
Specific Test =

Tiechman < Takayama < Spectroscopy

↓
Hemin

↓
Hemochromogen

↓
Brown-Rhombic Pink - Feathery



Test = species = Precipitin - Test (Blood, Hair, Bone)

2. Hair = Trichology

Scalp hair = 0.4 mm / day

Human	Animal → Coronal - Scale
Best = Precipitin - Test	
cortex ↑	medulla ↑
Pigment = periphery	pigment = centre
Medullary index = value ↓ < 0.3	↑ > 0.5

3. SEMEN

Sperm = 10%

S.fluid = 90%

Test

1. Barberios stain → Spermine → Picrate → Yellow - Needle [S - S]

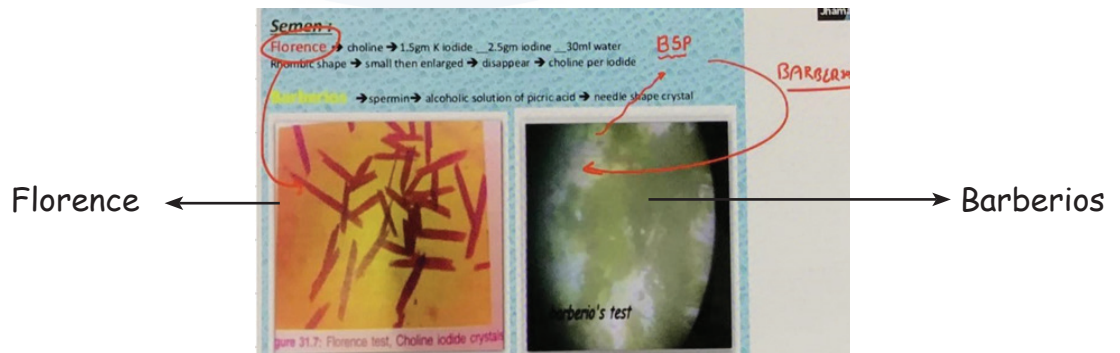
2. Florence → Choline → Brown - Rhombic → Kohli [C - C]

3. UV rays → Choline → White - Blue

4. Acid phosphatase → Sperm [Absent]

5. Creatinine phosphatase → OLD

6. LDH (lactate dehydrogenase test) M. Specific



Paternity / Maternity dispute = Affiliation - CASE



- a. DNA _____ (best) → Allec jeffrey
- b. HLA typing
- c. Blood grouping → best test → Elution - Test



Absent → CSF

LOCARD = French Criminalist

- 1. Poroscopy = skin → Pores = Identification
- 2. Exchange principle → Crime Scene Examination

Blood [Criminal <Material> crime scene] = finger-print



SEXUAL- OFFENCES AND PERVERSION

Chapter 06 :-

SEXUAL PERVERSION = Paraphillia

Abnormal act = Sexual - Gratification

Types:

1. Sadism = Pain → partner = Algolagnia
2. Masochism = ← Pain, Passive " "
3. Teacher → beating → student → Dippoldism = DANDA
4. Dead Body → necro → phagia = EAT Philia = Sex
5. Urine = Undinism
6. Feces = Coprophillia
7. Uranism = fore - play ← only
8. Sexual desire ↑ ↑ ↑
 - a) Male → Satyriasis
 - b) Female → Nymphomania
9. Child → Pedophile
10. Old → Gerontophile
11. Oral sex →
 - a) Male genitalia → Fellatio
 - b) Female genitalia → Cunni - Lingus → Tongue
12. Obscene telephone call → Scatologia
13. Obscene language → Capro - LALIA

14. Rub → FM → Frotteurism

→ CROWD

15. Wearing opposite sex clothes → Transvestism, Eonism

16. Opposite sex article = male → panty → Fetichism

17. Watching (nude girl/ girl sex) → Voyeurism, Scotophilia, Peeping-Tom

18. Exhibitionism = genitalia



Running naked = Steaking

Buttock = Mooning

Mastrubation = public space = Offence

19. Bobbit syndrome = FM → CUT → Penis

Sexual offence

Natural

[Peno-Vaginal]

Rape

Incest

1. sex with blood relation

2. No BNS

3. Not - Crime

Unnatural

[Other - Route]

1. Lesbianism = Sapphism, Tribadism [Stri]

2. Anal sex - Sodomy, Buggery (Buttock)

3. oral sex - Sin-of-Gomorrhoea

1,2,3 without consent / child <18yr

Adultery

Extramarital sex

Civil case = divorce

4. Bestiality = animal sex =

Sodomy

Adult (Pedarast) → child (Catamite)

RAPE

D → 63 BNS (Definition)

P → 64 BNS (Punishment) 10y → 6+4

Fm ₁	Fm ₂
↓	
Active	Passive
Butch	Femme
Dyke	

63 BNS → female



Vaginal

Anal

→ penis/penis like object/body-part

Oral

→ natural/unnatural

Urethral

1. Without Will

2. Without Consent

3. C ✓ = Fear

4. C ✓ = Fraud

5. C ✓ = Intoxicated / Insane

6. C ✓ = < 18y = Statutory - Rape

7. Not able to give C ✓ → Coma, Unconscious

Minimum age of boy below which incapable of sex → No-age-limit

Impotent → Not - Defence

P = 64 BNS → 6+4 = 10Y - L. I. (Life - imprisonment)

Examination



Accused

Consent = no need

Physical examination

Preserve = clothes, hair, saliva,
Blood

Smegma = glans

Penis → vaginal cells

↑ 4D



Lugol I2, Papanicolau stain

SAFE kit = Sexual Assault

Forensic Evidence Kit

Victim =

Consent = must =

→ Refuse = documented

= informed refusal,

female Dr.

→ Treatment = Injury, STD,
Pregnancy

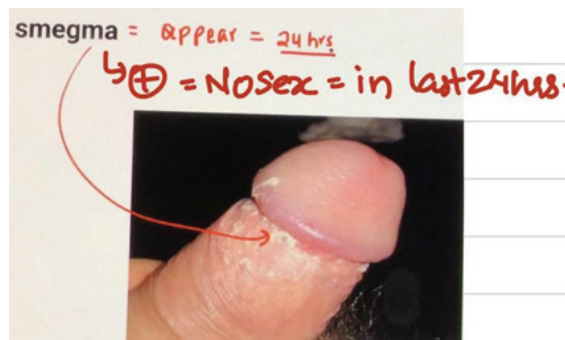
→ Ask her to remove cloth

→ Examⁿ lithotomy position

Preserve = cloth, hair, vaginal
swab, blood

Safe kit

↓
4



Child <18 years - POCSO = Act

Protection of child against sexual offence



COURT OF LAW

Chapter 07 :-

Cross Examination



Evidence → More valid → oral → document

Witness:

1. Eye witness → MC = Direct-evidence, Common wit = Common Knowledge

2. Hearsay witness → Indirect Evi, Not - Valid

3. Expert witness → Dr, finger print, fire arm, Chemical

1, 2, 3 → Dr. → All

4. Circumstantial Evidence → A → Fingerprint → Crime - Scene

5. Corroborative Evidence → A → Came out with knife → E seen

Cognizable offence

Def: Arrest - without - warrant

1. Rape → Kidnapping

2. murder → _____

3. ragging → Grievous - Hurt

4. Dowry death → _____

5. Hit & run = Death [Negligence]

Non Cognizable → with warrant

HURT, Fraud

COURTS:

President → Forgive → Amnesty

Commute → D.S.

1. Supreme court Appeal

2. High Court Confirm

Fast-Track 3. Session Court D.s.

4. Additional Session Court

5. Assistant sessions Court

6. Chief Judicial Magistrate court

Juvenile 7. 1st Class Judicial Magistrate court

8. 2nd Class Judicial Magistrate court

D.S. unlimited fine

10 yrs Un. Fine

7 yrs Unlimited Fine

3 yrs 50k

1 yrs 10k

Inquest = death

Investigation → COD I. O.



BNSS

Police

194 BNSS

MC

Unnatural deaths

minimum → H.constable

D.B

Crime scene → inquest report

2 Witness ↓

Panchnama

D.B → Dr._____ → P.M



I. O. ← DB



Police

Medi Colegal - Autopsy

Magistrate

196 BNSS

Six deaths

Asylum death

Beating - police

Custody - police

Dowry Death(<7y)

Exhumation → Dig out → DB

Firing - police

All police - Judicial Magistrate

all other - Executive magistrate



DM

SDM

ADM

Tahsildar

Active space

Not present in India;

1. Coroner inquest - 1999, Mumbai
2. Medical examiner inquest - USA, BEST^Q

Court Procedure for witness



1. Summon = Document

=time(10 AM)

date(25 Jan)

place

aka ↓

Subpoena



Under - Penalty

1
2
3

Civil	Criminal
→ No	yes
→	Both Criminal = higher
→	court Same level
	↓
	1 st received summon

2. Oath

→ Truth, complete truth, nothing but the truth

→ geneva

→ GOD →

- Not taken by → Atheist = I Solmnely Affirm

Age limit = <12yrs = No oath

false evidence → perjury

3. Examination of witness

a. Chief examination of witness → Same party lawyer

b. Cross examination of witness → Opp. party lawyer

c. Re examination of witness → Same party lawyer



Cross examination → toughest

→ no time limit

→ leading ques [ans → yes or no]



Was ram wearing white shirt?

Magistrate = Any point of time

4. Conduct money = Expenses → Witness,

1. Criminal Case → Not Paid

2. CIVIL CASE paid by → Party

3. Expert witness → criminal case paid by → Court

Dying Declaration India Oral, write, Victim Magistrate _____ OATH → No Dr - Cr. Ex → No Police - Any - _____ Valid = dies	Dying Deposition USA always by magistrate + lawyer yes yes (best)(protect the accused) superior
--	---

Active space

Age limit of evidence (Testify) → No - Age limit

Dumb witness: Write or Sign

Insane, intoxicated: Evidence Never Valid





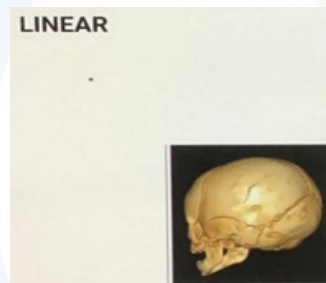
REGIONAL INJURY

Chapter 08 :-

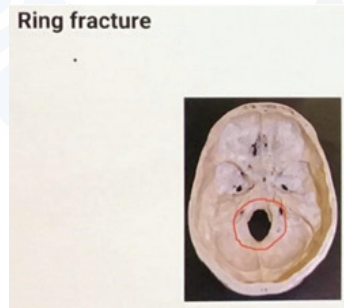
REGIONAL INJURY=

skull# MC Bone = Temporal

1. (MC) # fissured# ↓ aka



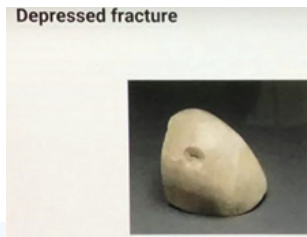
2. F. Magnum = _____ #



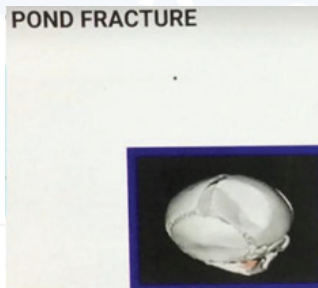
3. Tangential bullet = Gutter # = Oblique



4. Hammer → Depressed #
Signature #



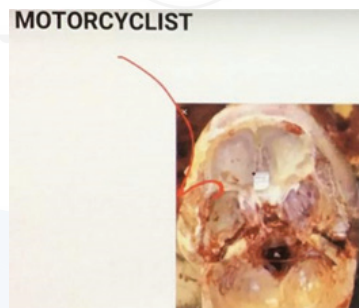
5. Infant → _____ # → Elastic Bone, Ping-Pong #



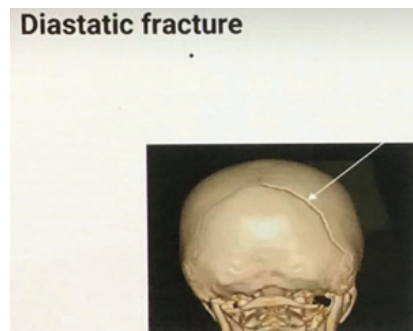
6. Motor cyclist #

a) Base of Skull # → A1/2-P1/2

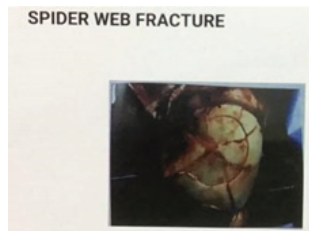
aka b) Hinge # type-1



7. Suture separated → _____ # M.C. = Saggital



8. Communitied# → _____#

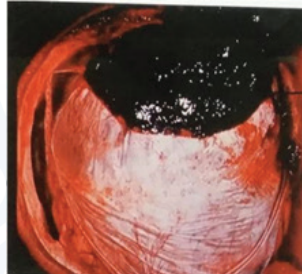


BRAIN:

1. Coup-counter coup injury = Mobile Head
2. Diffuse axonal injury → Axon broken → Retraction Bulb → Brain
3. Duret hemorrhage = Mid-Brain + Pons
4. Intracranial hemorrhage

1. EDH

- a). Exclusively - Traumatic, Pterion, MMA,
- b). Edal = Bi-Convex, Lucid Interval, Medical Neg



Uncon → Con ← Uncon

2. SDH →

Sickle shape

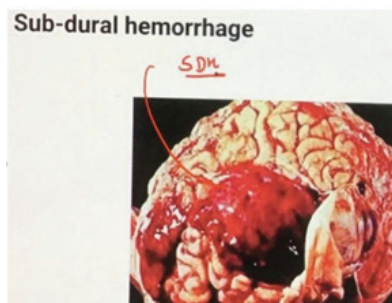
Concavo-Convex

Alcohol

Boxing, Banana, Bridging Veins

Child, Cross - Suture

Old



Active space

MIST

3. SAH → Ant. Communicating Artery, Aneurysm ← ↑ BP

a) M.C. Traumatic, Thunder Clap Headache, Star of Death

4. Intracerebral hemorrhage = Interparenchymal → MC Non traumatic

BP ↑ = Putamen

Vehicle accident

1. Seat belt = MC = Mesentery, Hollow Viscus, Patterned bruise

i. Chance# → Lumbar# Right Sh. → Driver, Lt. → Fellow

2. Dashboard = Patella#, P.D. = Hip → S. Nerve

3. Airbag = Face, eyes

4. Steering = Driver

i) Sternum #

ii) Rib #

iii) Ladder Rung Tear → D. Aorta

5. Glass(windshield) → Face = Sparrow - Feet - Mark

6. Neck = HF < HE → spinal cord → Whiplash Injury



Pedestrian Injury

1. Primary impact = leg = bumper# = tibia #
2. Secondary Impact injury = Glass, Bonnet
3. Tertiary Impact injury = hit ground = throw away

↓ aka

i. secondary injury`

ii. Most dangerous

Multiple grazed abrasion, tyre mark

Blast Injury

1. Primary blast injury → Air - compression → T. M. > Lung
2. Secondary blast injury → [abrasion / contusion/punctate laceration] → Marshall triad, [flying missile → Body]
3. Tertiary blast injury → you = ground = Skeletal #
4. Quaternary blast injury → (heat, burn, flame)

Blunt trauma = MC organ = Spleen

Penetrating trauma = MC organ = Liver

Blast injury = MC organ = T. M.

Underwater blast = MC organ = Head - above = Int, [colon]
 - Below = T. M.



THERMAL INJURY

Chapter 09 :-

Hypothermia ($< 35^{\circ}$ Celsius)

1. Moist cold $\rightarrow 2-8^{\circ} \rightarrow$ Trench foot



2. Dry cold $\rightarrow -2.5^{\circ} \text{C} \rightarrow$ frost bite = nose



Death = PM findings

PM staining $\rightarrow$ Bright - Pink

Stomach $\rightarrow$ _____ $\rightarrow$ _____



Mucosa

↓
Black = Spot

↓
Wischnewski - Spot



Hyperthermia (temp ↑ = _____)

1. Heat cramps => Sweating ↑ = Water => Out

Miners cramps => _____ Muscle - Cramp
Fireman cramps

2. Heat stroke → Sun-light, Humid, Fever, Infection, Muscle Activity ↑

Sun stroke => _____ => _____ => sweating stopped

Heat Hyperpyrexia _____ (>40.5° C, >104°F)
Thermic fever

3. Heat syncope

Heat exhaustion _____ = Sweating ↑ ↑ ↑ ↑

Heat prostration _____ = Venous - Return ↓

_____ = Cerebral blood Flow ↓

= Vascular - Collapse

MIST

BURN**1. Flame = Dry = above contact**

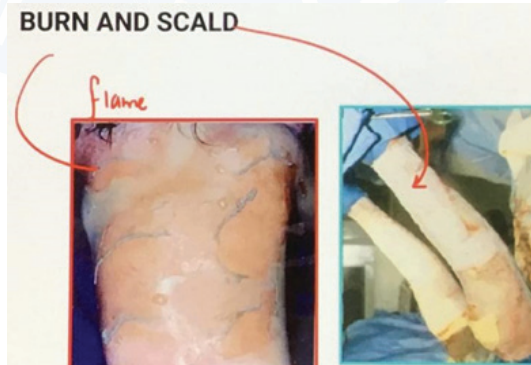
- a) Feel = 44°C
- b) Blackening → Skin
- c) Burn → Singeing of hair = Skin
- d) Red line (+), Blister (+)

2. Scald = Moist - Burn = below contact

- a) Liquid / steam
 - ↓
- b) > 60°C Celsius
 - Red line (+)
 - Blister (+)
- c) Bleaching (+)
- d) Soddening [moist] (+)

3. Chemical Burn → Below

- ↓
- ⇒ Red line (-)
- ⇒ Blister (-)
- ⇒ Ulcer (+)

BURN AND SCALD

Classification = Burn

Dupuytren → 6 Wilson → 3

1°	↓
2°	1° = Epidermal
3°	
4°	2° = Dermo-epidermal = M. Painful
5°	
6°	3° = Deep = pain ↓ = Nerve - Ending → Lost

MC COD = Burn

Immediate => Asphyxia

Early => Hypovol. Shock

Late => Septic Shock

Overall => Septic Shock



	AM Burn	PM Burn
Redline	Present	Absent
Vital Rxn	Present	Absent
Enzymatic Rxn	Present	Absent
Blister	+ve [albumin + Cl]	+ve [air + thin fluid]
Smoke →	CO↑ = Blood	Normal
↓ Carbon Particle, Soot P.	Trachea (+) ↓ M. Ch.	(-)

Non-specific finding

↓
Burn

↓
AM + PM

1. Heat- hematoma = brain

Look like = EDH

MIST

= chocolate-brown, Honey - CombApp.

2. Thermal# = skull#

a) _____

b) Dehydration

3. Heat rupture → skin, joints, extensors, fleshy area



Dehydration

Margin => Irregular



Look like



Laceration

4. Pugilistic attitude

a) Boxing attitude = Shrink due to Cogulation of muscle Protein

b) Fencing attitude

c) Heat Stiffening ($> 65^{\circ}\text{C}$)

Heat Rupture



Base = Pale
B.V. = Intact

Pugilistic attitude



Look like



Rigor - Mortis

Body surface area = Burn

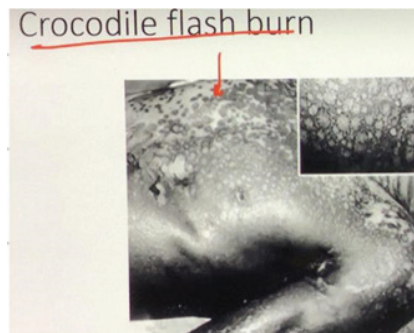
1. Rule of palm → 1% BSA
2. Clinical practice
 - a) Lund & Browder chart
 - b) (Head, Thigh, Leg = 40%)
 - c) Remaining = 60%
3. Wallace rule = Rule of 9
 - a) $11 \times 9 = 99\%$
 - b) 1% = genitalia

Electrocution → (1) Joule - Burn

Entry burn = Endogenous burn = Low-Voltage, Firm Contact



2. High voltage = Loose Contact



3. High Voltage + air gap = spark / Flash

PM CHANGES Rigor - Mortis = Early

→ Molten metal = skin, soft tissue → Current pearl

→ Molten Ca^{2+} / Phosphate = Bone



= Bone - pearl



= Wax - Dripping

Lightning = _____

1. Shoulder, flank
2. Filigree Burn
3. Arboroscent - Burn
4. Lichtenberg - Burn
5. Kiran → = keranographic burn

Pink

- Erythematous
- Branches of leaf
- Denaturation → RBC
- Air Compression → Front
- Blood → Vessel → Does not correspond



BNS →

Hurt → 114 BNS

Grievous - Hurt → 116 BNS



1. Emasculation = Erection of Penis
2. Eye sight → Lost

3. Hearing → Lost → Sx → recovered = Grievous hurt
4. Privation of joint / member → Part = Independent - Function
 - a) Nail = lost = Hurt
5. Impaired, loss of power
6. Permanent dis-figuration to face or head = Acid - Attack
7. # / dislocation to bone or teeth
8. (a) Dangerous injury = life danger
 - STAB = Knife
 - Bullet = Body
 - Head Injury
- (b) Severe painful $\geq 15D$
- (c) Not able to follow routine work $\geq 15D$

Hospitalized = 16D = not G.H

Dangerous weapon

1. Stab , poison , animal
2. Cut, blast
3. Shoot, burn

Torture =

MC = beating → sole → Falanga, Bestinado, Foot, Boot

→ ear → Telefona

Long standing = Planton

Picana = electric torture = [genitalia → cattle prod]

Face → suffocation = dry - submarine

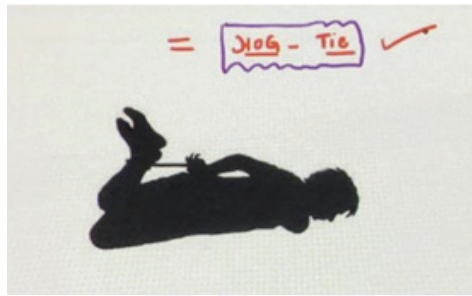


Plastic - Bag

Face → Water - Wet - Submarine

MIST

Chilli paste → anal canal → Bellary → Burn
 Hot rod → anal → Black - Slave



Strappado / La-bandera



Pole → Parrot-Perch
 = Saw-Horse = Pointed
 Crush = Belana
 Bamboo = Chepuwa

Crucifixion



Jackknife position

= Positional Asphyxia



Positional Asphyxia



MECHANICAL INJURIES

Chapter 10 :-

① Abrasion = epidermis - Loss

a) Scratch → Nail, Thorn, Pin _____ Linear

b) Grazed → sliding Ab

aka {
i. MC = RTA
ii. Gravel - RASH
iii. Brush - Burn



c) Pressure abrasion

= bite mark, ligature mark = hanging

d) Imprint Ab [Patterned] = tyre mark



d) Age = scab = dry blood = 12-24 hrs, Bright-Red 2-3D

e) Scab → brown [4-5D] → fall [7D]

② Contusion = bruise → Blunt Margin = IR Regular

Color changes = dermis = B.V. = Rupture

1. Red = Hb (Fresh)
2. V or I = blue = deoxy Hb (Few hrs → 3D)
3. Brown = hemosiderin (4D)
4. Green = Biliverdin..... (5-6D)
5. Yellow = bilirubin (7-13D)
6. Original color = no pigments (14D)

Artificial bruise

Plant rub = PSM

PLUMBAGO

Semicarpus

Madar (Akdo, Calotropis)

= _____

= _____

Margin = Regular

= Itching (+)

= Blister (+)

= Chemical Rxn (+), Brown

2. Patterned bruise

Lathi / rod → tramline/rail, throttling → Bruise → Six - penny → Coin

Beating baby → pinching = butterfly bruise

3. Ectopic bruise/ migratory bruise

	Trauma	Bruise
Racoon sign	ACF #	Peri-ocular
Battle sign	MCF #	Mastoid tip



③ → Incision → Sharp edged = weapon = knife [Cut/Slice/Length= Max.]



End → Tailing

1. Parallel = skin

2. Margin = sharp

a) Clean cut (Regular)

Knife with irregular margin at Axilla & Scrotum



Irregular margin looking incision = Laceration looking incision

Seen on (Magnifying - Lens)

a) Cut tissue

b) Cut hair bulb

Fibroblast = 48-72 hrs^Q

Collagen fibers = 3-5 days^Q

Cut = genitalia = Homicidal

Cut = wrist, Neck, fore-arm → Flexor

1. Multiple = Hesitational Cut

2. Superficial = Tentative Cut, Self Inflicted

3. Suicidal = Trial cut, Intentional Cut

4. STAB → Sharp edged weapon = knife =



Perpendicular → Stab [max dimension] = Depth

1. Skin → Puncture
2. Cavity → Penetrating = Liver
3. Through and Through → Perforation

Shape → skin lines = lines of Langers = Collagen - Fibers

Fish tailing →  Tail → Blunt, Single - Edged - Weapon

Japan = suicide = STAB → SWORD → ABDOMEN = Card-Collapse



Harakiri/Seppuku

5. Laceration, Rupture, Irregular Margin, Swallow - tail

Heavy blunt force → hammer, stone

= dmg to E + D + S.C.T.

Magnifying lens → Crushed Hair Bulb & Tissue = Tissue - Bridges

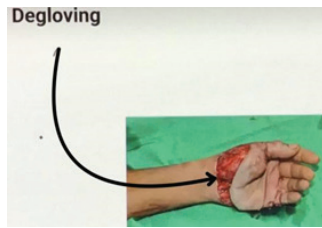
Types:

1. Avulsion → Rotational force
→ Run-over Injury
aka → Shearing Laceration

A. Flaying → Flap

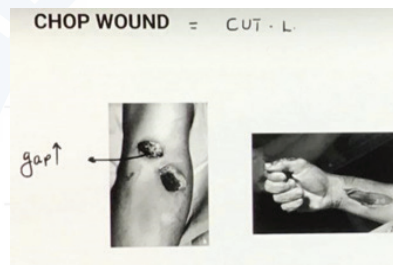


B. Degloving



2. Cut - laceration = axe = Heavy Sharp edged weapon

- a) Cut tissue + crushed tissue ≠ Bone
- b) Gaping ↑ → Chop



3. Tear = Door - Handle [M. C.]

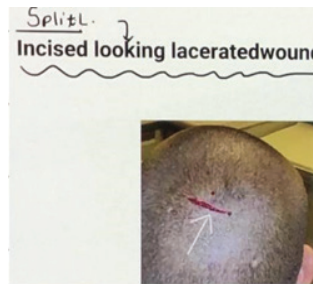
4. Stretch L ● → Stretching of skin → Flap
→ Tangential - Force

5. Split L ● → bony prominence = skin

- a) SCALP
- b) Fore-head,
- c) Eye - brow, Chin
- d) Post ● - elbow
- e) iliac crest
- f) Shin

→ Hard - Surface
Skin → Close = Bone
→ Regular Margin

aka = Incised looking lacerated wound



Defence wounds

1. MC → Active = Palmar Surface → 1st - web - Space

2. Passive → Extensor = Fore-arm, Dorsum = Hand, Medial / Ulnar
= Fore-arm

Fabricated wound:

1. You → your body = self-inflicted

2. Other → consent = self-suffered



THANATOLOGY

Chapter 11 :-

THANATOLOGY

→ Study of death

= sydney

2 types

1. Somatic death

a) Lung, Heart, Brain: stops → Bishop tripod, of life

2. Cellular death

a) Cell dies → molecular death

Between somatic and molecular death lies supra vital interval during which organ harvesting is done.

Cornea → 6hrs

Bichat mode of death:

1. Asphyxia

2. Syncope

3. Coma

Brain stem death = Minneosta - Criteria

Atria - Mortis aka → Gateway of Death

↓ ↓ ↓
Respⁿ Cirⁿ Brain Funⁿ

Any- one - Stop

↓
Somatic-Death

Active space

Suspended animation = CPR

(Apparent Death)

Alive but dead looking

Involuntary Causes

→ Anesthesia

→ Born - new, barbiturate

→ Cholera, concussion = brain injury

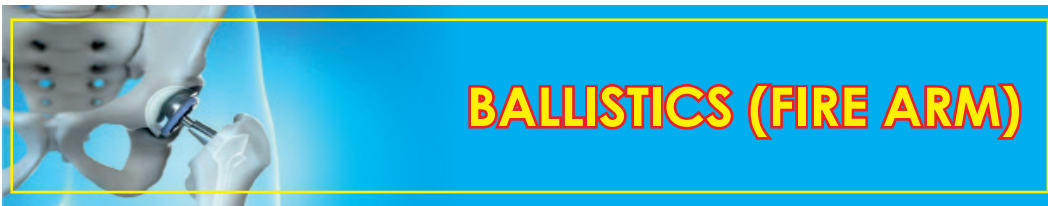
→ Drowning

Electrocution

Hypothermia, hyperthermia

Sun stroke, shock, insane

→ Trance [yoga] = voluntary



Chapter 12 :-

Father of ballistics: Calvin Goddard

Two types of weapon

Rifled → Land, Groove (Rifling)	Smooth Bore
Calibre b/w = Two-lands	Bore
Bullet = 1/1, lead	Pellets = multiple = Lead
Velocity: ↑↑	↓
Rotation: +	-
Range: ↑	↓
Revolver = 200 yards Pistol = 400 y Military rifle = 1000-3000y	Shotgun = 50 y range↑: 1. Choking → Dispersion of Pellets ↓ 2. Paradox → Muzzle end → Rifled ↑ bore lead pellets - Size ↓ 454gm
12 B 1 /12 24 B 1/24	

Gun powder(GP) → burn → smoke → P ↑ → Bullet / Pellets

MIST

1. Black → Pocso

- a) Pot. Nitrate = 75%
- b) Charchol = 15%
- c) So → Sulfur = 10%

2. Smokeless = better

- a) Single Base = Nitrocellulose
- b) Double Base = Nitrocellulose + Nitroglycerine
- c) Triple Base = Nitrocellulose + Nitroglycerine + Nitroguanidine

Test to detect GP: → Hands

- 1. Harrisons
- 2. Atomic absorption spectroscopy
- 3. Neutron activation analysis
- 4. Dermal nitrate test = paraffin test → not used^Q
- 5. Scanning electron microscopy → MC^Q

Bullet = Lead → crime scene → pick with Hands, (gloves)

Bullet

1. Primary marking

- a) Class = Gross
- b) = Rifling
 - i. _____
 - ii. _____

2. Secondary marking

- a). Individual
 - i. Irregular Barrel-Surface → Microscopic
- 1⁰ & 2⁰ → bullet fingerprinting

Types of bullet

- 1. Dum dum bullet → more dmg → Chiesled → Explosion
- 2. Tracer bullet → Path → Trace → Powder

3. Express bullet = ↑ velocity → Hole → Base

4. Incendiary bullet = phosphorous

5. Frangible bullet → Fragments

6. Souvenir bullet → Skin - S.C.T = Lead - Poison (Chronic)

7. Glancing bullet → Gutter#

8. Tumbling bullet → Tumble

9. Yawning bullet → Slow/Irregular entry wound of yawning bullet = Key-Hole

10. Tandem bullet = $O \rightarrow O \Rightarrow \textcircled{\text{e e}} = \text{Two} \Rightarrow \text{Piggyback}$

11. Ricochet bullet = Deviated → entry wound features are absent, 30°

	Entry wound	Exit
Size	Small	Large
Contact shot	Large, Stellate	Small
Margin	Inverted	Everted
Tissue color	CO + Hb → cherry red	-
Expulsion of tissue	-	+
Abrasion collar	+	-
Grease collar		
SBT	+	-
S → Singeing of hair = flame		
B → Blackening = smoke of G.P		
T → Tatting = un-burnt G.P.		
Contact = large = SBT (-) In-Track →	Stellate	-
Close = SBT → (+)	Circular	-
Near BT/TT (+)	Circular	-
Distant = small = SBT (-)	Circular	-
Shotgun= pellets		
Shape		
Contact →	Stellate	-
Oval	<1 mt	-
Rat-hole	1-2 mt	-
Satellite	2-4 mt	-
Individual pellets →	>4 mt	-

KENNEDY PHENOMENON

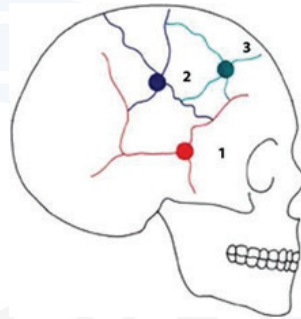
a) Alteration or surgery to gunshot's entry or exit wound upon PM it is difficult to identify the entry and exit

b) Rayalseema → effect → STAB → Bullet → Implant

c) Pueppe rule → Skull → # Line → Sequence of bullet

→ 1ST Shot = Natural

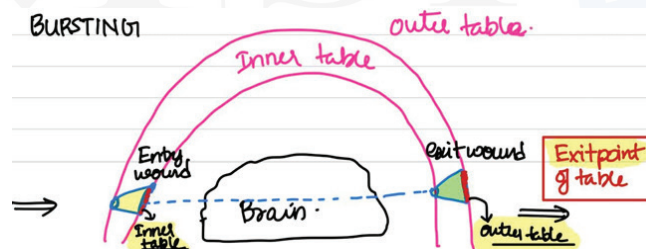
→ Subsequent Shot → Stop



d) Beveling of skull → bursting

Seen

→ Exit Point





Chapter 13 :-

Asphyxial death $O_2 \downarrow$

→ dies

→ CPC triad

→ C = Cyanosis → blue

→ P = Petechiae

→ C = Congestion = Organ → Red

Petechiae → Tardieu-Spot = eye, skin, mucous membrane →
compression of neck and chest

Jugular V. → 2 kg

Carotid A. → 5 kg

Trachea → 15 kg

Vertebral Artery → 20-30 kg

1. Hanging

Def: Suspension of Body + Neck-Compression

Suicidal > homicidal, accidental, judicial

= MC method of suicide in India

Painless, Fatal period = 3-5 min

Types =

① Complete = → No Part Touching Ground = COD = Asphyxia → 15kg

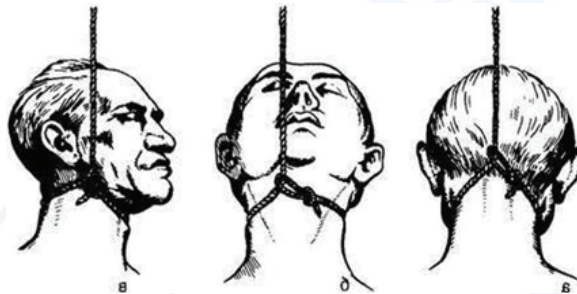
② Incomplete → Partial → Some part → Touching ground

COD = Venous Congestion



③ Typical = knot → Occiput

④ Atypical = knot → Any-Other-Place



MC COD = Asphyxia + Venous Congestion

PM Changes → CPC +ve

→ tardieu spot +

→ protrusion of tongue +/-

→ abduction # → Hyoid bone# 10-15% (A-P#)

→ M. Specific → ligature mark → Neck

Incomplete

↑ (-)

KNOT

Pressure abrasion

Above Thyroid* White, Yellow, Parchment

Dissection → Pale glistening

M.char = A.M-hanging ① Dribbling of Saliva = Opp. Side of Knot

= death due to hanging → Pterygopalatine Gang.,
Submandibular gland

② = LA Facie Sympathetique → Same side KNOT=eye-open



Cervical sympathetic chain

Pupil = Dilated

- b. Homicidal hanging → Lynching → _____ = N.america
- c. Sexual perversion [masochism,transvestism] → accidentally hanged
→ Sexual - Asphyxia aka Auto-erotic Asphyxia aka Kotzwainism

d. Judicial hanging

→ India → atypical

Complete hanging → sudden drop → carotid tear → Amussat sign

Hangmans knot → Location of knot: Angle of mandible (Ideal / Best = Below chin)

Hangman# → C₂ bone #

Dislocation of C₂ = Brain - Stem Damage

2. Strangulation

Only compression of neck

Homicidal > accidental, judicial, suicidal

- a. → ligature strangulation
- b. → manual strangulation: Hand = Throttling → always homicidal
- c. → elbow strangulation: Fore-arm = Mugging
- d. → Stick strangulation: Bansdola
- e. → judicial strangulation Garoting
→ spain, turkey
→ spanish, windlass

PM finding = Ligature strangulation

= cyanosis, petechiae & congestion = more prominent than hanging

= tardieu spot +ve

= Resistance Injuries

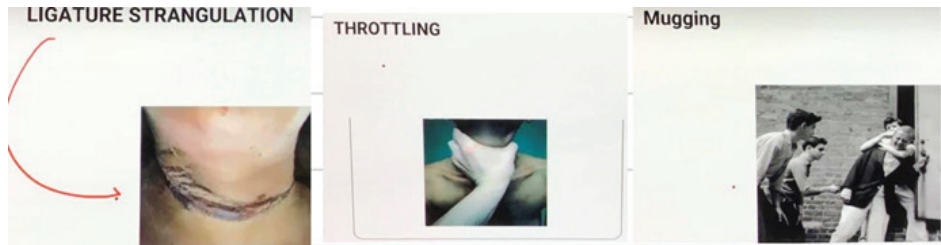


Body

= Thyroid bone# = MC

Ligature - Mark is complete, (+) → Peri-ligature injuries, mark transverse, (at thyroid or below thyroid cartilage)

Dissection = hematoma / hemorrhage



Throttling

It is always Homicidal

Cricoid # - Specific

Findings:

Thyroid bone# [MC in ligature strangulation]

Hyoid bone# = Adduction # 50% (Inward - Compression)

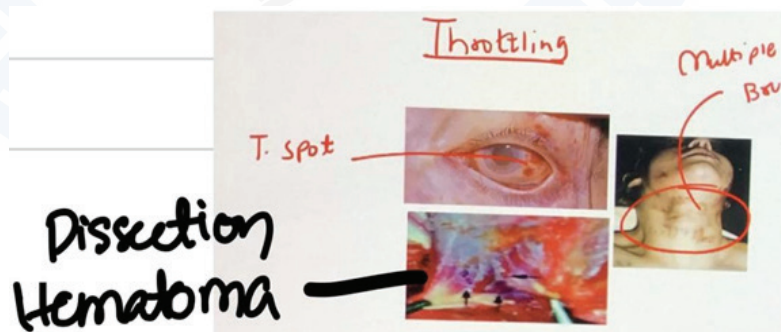
Neck → Multiple bruises, Six penny bruises

Bruises → dissect = hematoma/ hemorrhage

→ fingernail abrasions Semilunar, Crescentric

PM findings = CPC (+)

= _____



3. Suffocation → No Suspension, No Compression [Neck]

Def: Purest form of Asphyxia

COD → Always - Asphyxia

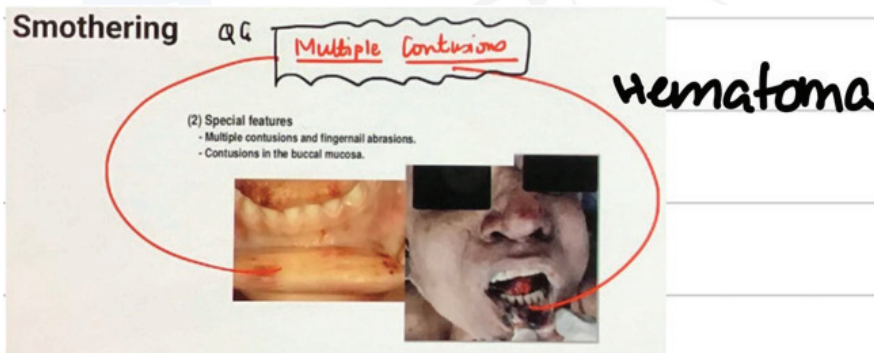
CPC → Always - Present

Types:

1. Environmental $O_2 \downarrow \rightarrow CO$
2. Food $\rightarrow$ trachea $\rightarrow$ choking = (A)
3. Cloth $\rightarrow$ oropharynx $\rightarrow$ gagging $\rightarrow$ (H)
4. Mouth + nose = blocked = smothering $\rightarrow$ (H) (A)
5. Trauma $\rightarrow$ chest fixed $\rightarrow$ traumatic asphyxia Railway, RTA Stampede, Earthq. [perthe syndrome]
6. 4+5 = Burking = homicidal
7. Cafe coronary = rogger hougen $\rightarrow$ Heart
 Food = Bolus $\rightarrow$ larynx/pharynx = block Into xicated
 = mechanical asphyxia $\rightarrow$ vagal inhibition $\rightarrow$ C. Arrest
8. Overlaying $\rightarrow$ (A) Mother $\rightarrow$ Child

Smothering

Multiple contusions = Peri - Oral $\rightarrow$ Fingernail - abrasions



4. Drowning

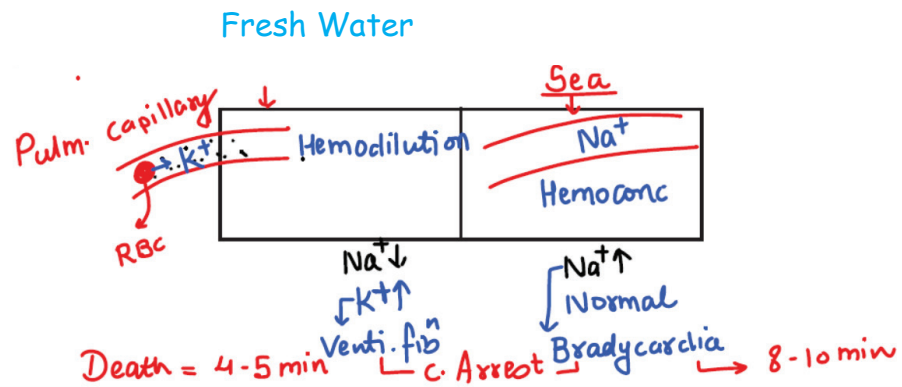
Body $\rightarrow$ water

Accidental [MC] > suicidal (F.P = 5-8 min), homicidal

Types

1. Typical: wet drowning : water $\rightarrow$ lungs = Less = Loss of Pulm. Surfactant, More = Cardiac Arrest

a) Fresh water: mechanism: _____



b) Sea water: mechanism: _____

2. Atypical

a) Dry: no water in lungs, COD = Laryn gospasm

b) Cold water:

↓
Nose, Ear, Epigastrium = Vagal (-) = [C. Arrest]

Mechanism: Shivering

i. aka:

1. Submersion Synd.
2. Hydrocution^{QQ}
3. Immerssion Synd.

c) Delayed: Delayed complications [1-2D → Death]

- i. Pneumonia
- ii. Metabolic acidosis
- iii. Electrolyte imbalance
- iv. Encephalopathy
- v. aka:

1. 2^o Drowning
2. Near Drowning
3. Post - immersion-Synd.

PM changes:

AM drowning → death in water

1. Most char.: Cadaveric - Spasm
2. Water+Mud → G.I.T.
3. Water → lung = → Heavy → Emphysema - Aquosum
 - a) Water+mucus = Froth-Mucoid water → Leathery → Resp. (+)
 - b) Mud particle = Trachea
 - c) Sub pleural hemo = Paltauf Hem.
 - d) Mucoid Froth from in nose & mouth
4. Gettler test → Cl^- in Heart

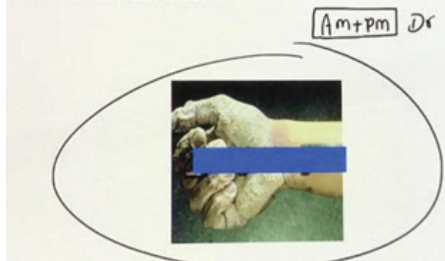
↑ ↑ ↑

(Get → Close → Heart
5. Water=algae=Diatom → Organ → ___ < → bones=Femur > Sternum
 Diatom → Unicellular & Silica → Acid-digestion - test

Non-specific findings: = AM + PM

1. Washer woman hands
 - a) skin → Palm-Sole
 - b) Wrinkle → 3-hrs.
 - c) Bleaching → 12-hrs. - 24hrs
 - d) Soddening → 12hrs.
 - e) Cuticle-peeled off → 3-4 D

Washer woman hand



2. Flotation

a) Decomposition = gas ↑

i. Summer = 24 hrs

ii. Winter = 1-2 day



MEDICAL JURISPRUDENCE

Chapter 14 :-

MEDICAL JURISPRUDENCE

Doctor → practice = laws

NMC → MCI act = 1956

Schedule I = MBBS → india

Schedule II = FMG

Infamous Conduct → Incomplete - list 6A

Alcoholism

Addiction

Adultery

Abortion - Criminal

Advertisement - False

Association → Comission

BRIBE

Covering → Non-Medico

Dichotomy → Fee Split → Pt. giving Comission

Euthanasia →

False - Certificate

Gift →

1. MCI: Registration → _____

2. SMC (state medical council):

Dr=infamous conduct aka professional misconduct

1. Notice

2. Erasure of name of Dr[medical register]

Professional death sentence / Penal-erasure

3. Appeal → Health-ministry

Professional secrecy → exception[privileged communication]

1. Crime → Police / Mag.

2. Inf. disease → Relative

3. Veneral disease → STD = Partner

4. Interest of patient → Psychotic → Suicidal → Relative

5. Court of law ← Dr.

6. Servant disease → Employer [Train driver → Color-blind]

7. Interest of self

↓

[Dr. → Sx. → Newspaper] → Pt. → Consent

8.a) negligence → Relative / Pt

b) Notifiable disease → Cholera, TB, Plague = Health Deptt.

Medical negligence

Malpraxis

Civil = TORT	Criminal
Mild damage No care No skill Civil, Consumer Court	Gross negligence Gross damage/death
Money	Money + 106 BNS = 2y± fine
Burden of proof → patient	Doctor
Ex: Wrong Prescription ↓ Card S.E.	Ex: Sx → Swab → Body Rt leg Gangrene → Lt Ram-lal → Shyam-Lal Alcohol → Sx → Artery → Collapse

Anything/fact → sufficient to prove negligence → RES - IPSA -
 ↓
 LOQUITOR →

speaks for itself

Defence for doctor

1. No duty
2. Informed consent → patient → Sx
3. Time limitation → 2y [Res - Indicata]
4. Res-Judicata → Civil
5. Contributory negligence → [Dr. +Pt.] → Civil → burden of proof=Dr
6. Therapeutic misadventure = Medical - Maloccurrence
 - a) T/t → Unintentional, Accidental, Complication

Burden of proof = Dr. Civil + Criminal

Vicarious liability → Let Master - Answer, Respondent - Superior

Fortis = Employer → Dr. → Negligence

Euthanasia

= painless death → Mercy - Killing

A. Active

a) Drug =

i. Phenobarbitone → Barbiturate

ii. Thiopentone

iii. Opium

iv K⁺

B. Passive → Remove -Life line → ICU, Ventil

a) India → Passive → _____

1st country → Netherland

MIST

Other countries ex: Belgium, Sweden, Thai

Consent → Dr & patient [Indian - Contract - Act]

Types:

1. Expressed: Verbal, Written
2. Implied : Gesture, Act = Dr. → BP → Pt.
3. Substituted: Relative
4. Informed : → Written = Best

Exception = life threatening condition / Emergency



RTA → C = (X) → 30 BNS

5. Open/ blind/ blanket = Not - Valid = INDIA

= Valid → = USA

Single C. = Multiple Procedure

cholecystectomy → organ problem → remove

LAWS:

1. For surgery] ≥ 18y → Consent

Guardian



2. (<12y, Insane = consent = never valid
3. General physical examination → >12yr → consent

Medical ethics → Moral - Principle = Pt.

Medical etiquette → Courtsey → b/w, Medical → Professionals



IDENTIFICATION

Chapter 15 :-

Corpus - Delicti → Body of offence

Race = cephalic index = skull

$(\text{Breadth of skull} / \text{length of skull}) \times 100$

Race	Value
Dolicocephalic Negro, Pure-Aryan	70-74.9
Mesaticephalic Indian China Europ.	75-79.9 → Masti
Brachycephalic Mongol	80-84.9

Caucasian: teeth = carbelli cusp,

Mongols: teeth = shovel shaped incisor, hair=medulla→Round→ Mongol

ORBIT = Round = Mongol, Triangular → Caucasian, Square → Negro

Sex = bar body → female = Buccal - Mucosa

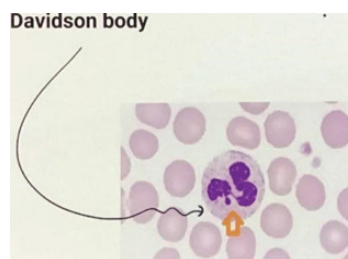
Better = Davidson body = FM

↓ = 6% neutrophils

Drum - Stick

↓

Cl. tetani



Sex → bone → Krogman table

100 % → Complete - Skeleton

95 % → Pelvis -

90 % → Skull -

80% → Long-bones

} → 98%

Male(skeleton)	Female(skeleton)
Large, heavy	↓
Prominence: More Prominent	Prominence: frontal, Parietal Emi
Sternum length: _____ ↑ (ashley rule)	_____ ↓
Pelvic inlet shape: Heart	Round
Obturator foramen: Oval	Triangular
Angle: ↓ V	Angle: ↑ U Suprapubic angle: > 90° Mandible : > 125°
Chielotic line: Sacral ↑	Chielotic line: _____ = pelvis ↑
Preauricular sulcus: Rare (-)	Preauricular sulcus: + _____ ; pelvis + : 3° sexual ch
Bony index value: ↓ ↓ ↓ Sacrum → (Corpobasal Index ↑)	Bony index value: ↑ ↑ ↑ Best= Sciatic Notch Index
	Best parameter → Greater-Sciatic Notch ↓ Wide, large, shallow
Ischial Tuberosity Inverted	Everted

Bone = Stature

=Length * [Multiplication - Factor → Karl - Pearson]

↓
Femur → 3.7 [+0.8]

Tibia → 4.5

Humerus → 5.3

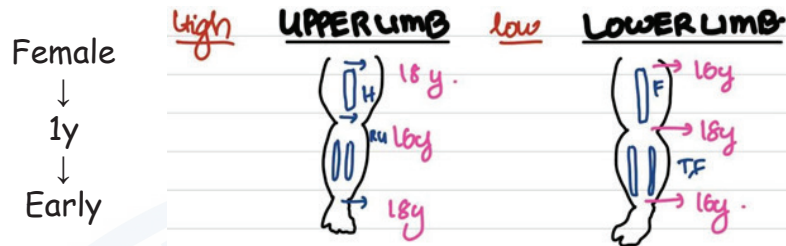
Ulna → 6.1

Radius → 6.5 [+0.4]

Carpal bones

1st = Capitate = 2-3 months 2nd = Hamate

Last = Pisiform = 8-12 y



Skull = 1st suture = Metopic = 9-12 months

Last suture = Squamo - Temporal >80y

Best → Sagittal → 50-60y

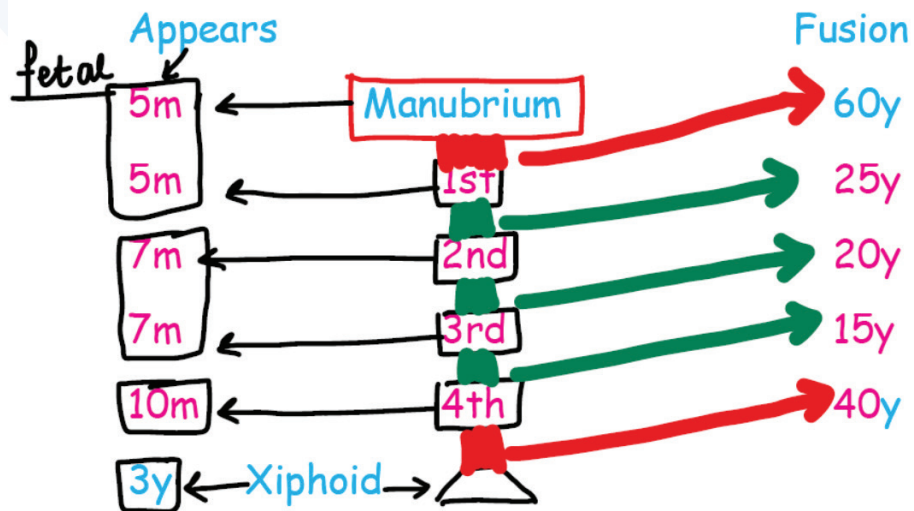
Coronal → 30-50y

X-ray: lateral skia-gram Pterion = 65y

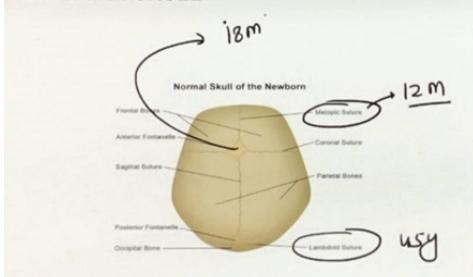
Lamboid → 45y

Sacrum = single = 20-25y

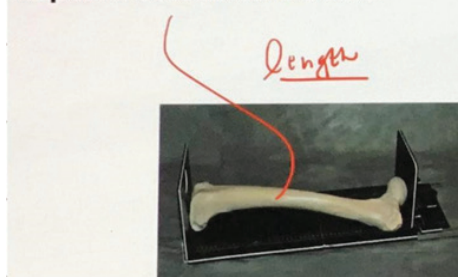
Sternum:



NEW BORN SKULL



Hepburn osteometric board



Bone

Teeth = dentition = (Age = Child)

T = temporary teeth = 20

↓
Milk-teeth

↓
Deciduous teeth

P = permanent teeth = 32

T = 20 no. Of teeth

I = 8

C = 4

PM = 0

M = 8

				Maxillary							
				I	C	PM	M				
RU	2	0	1	2	2	1	0	2	LU		
RL	2	0	1	2	2	1	0	2	LL		
				Mandibular							

Sequence of eruption → T

→ 6m I → _____ = 6m → Lower Central (I)

→ 12m M →

→ 18m C →

→ 24m M → _____ $2-2.5y = 20T$

↓
6y-7y

↓ ↑ P+T ↓ → mixed dentition → 24

_____ 11y - 12y

Active space

Sequence of eruption - (P) = 32

4 Mummy 6-7y → 4 P+20T

4 Is 7-8y → 8 P+16T

4 In 8-9y → 12P+12T

4 Pain 9-10y → 16P+8T

4 Papa (10y) 10-11y → 20P+4T

4 Can 11-12y → 24P

4 Make 12-13y → 28P

4 Medicine 18-25y → Wisdom Tooth = (32P) → Successional → 20

→ Super added → 12M

				←				→			

4. Pic of tooth [anatomical chart formula]

5. FDI → Federation dental internationale

FDI		Modified FDI	
Rt	①	1	2
18	11	18	11
48	41	38	31
21, 22, 23	21	21	28
31	31	41	48
38	38	3	4
41	41		
48	48		

Identification method

1. Poroscopy = Pores = Skin = Locard

2. Chelioscopy = Lip-Print = Suzuki

3. Palatography = Hard - Palate = A 1/3, Rugoscopy = Allen

4. Superimposition technique → Confirm → DNA

Skull → Photo = Screening

5. Tattoo marks = _____

↓ ↓ ↓

Dye

↓

Lymph. Node

↓

Regional - L.N. Biopsy

Live _____ (Old = Infra - Red)

Dead _____ = (Faded = UV- Ray)

Decomposed _____ = _____

6. Anthropometry = Bertillion system, Portrait - Parle

Identify → Body - Measurement

Body - Marks

Descriptive-Data + 4 - Photo

7. Best identification method → Criminal = Fingerprint →

Dactylography → [Galton System]

MIST

Discovered → W. Herschel

1st Bureau → Kolkata

Fetus → 12wk - 24 wk

Identical twins → Not Same = Quetlet Rule

Permanently lost → leprosy
Radiation
Charring = Burn

dye → I_2 , $AgNO_3$

Osmium Tetra Oxide

Minimum point criteria = similar = 10- 12 (Depth = 0.6mm)

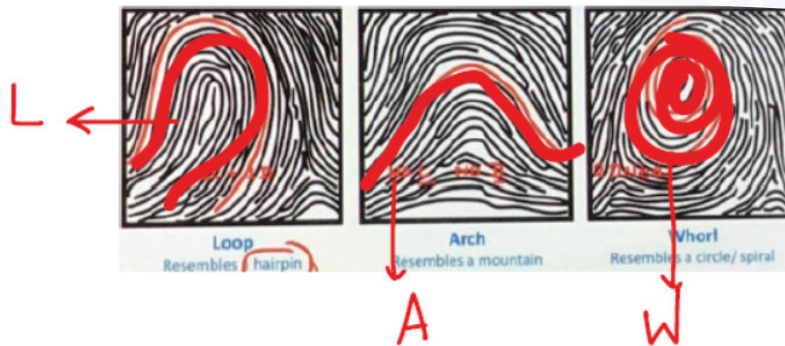
Galton - Henry classification =

Love Loop 70% = MC  Hair - Pin

Wife Whorl  = Circle

and Arch  Mountain

Child Composite = Mixed = Least





TAPHONOMY

Chapter 16 :-

TAPHONOMY = PM changes

1. Immediate changes

1. Insensibility / Loss of Vol. Power (Earliest)
2. Resp (-)
3. Circulⁿ (-)

2. Early changes (all remaining changes)

3. Late changes

1. Decomposition [Putrefaction]



2. Types [Skeletonization, mummification, adipocere]

Early changes:

1. Eye →

a) 1st → Kevorkian - Sign | Rail - Road

Retinal vessel → segmentation

b) 2nd → K⁺ ↑ → in vitreous humor

c) 3rd → sclera change = Tache - Noire (3-6hrs)

2. Temp ↓ → = Algor - Mortis

a) Ideal → Rectum place → _____ celsius

b) Rate of fall

i Summer → 0.75° F/hr

0.4° C/hr

ii Winter → 1.5° F/hr

0.6° C/hr

MIST

c) Exception = temp ↑ → = 1-2 hr. → PM-Caloricity

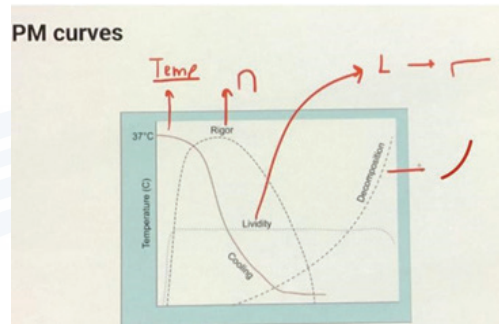
S i. → Sun-Stroke, Pontine Hem.

S ii. → Septicemia, Cholera

S iii. → Strychnine Poison, Tetanus

Sigmoid-curve

Inverted 'S'



3. Postmortem staining = Suggilation Livor-Mortis

- a) Postmortem lividity → Skin - Staining
- b) Hypostasis
- c) Vibices
- d) Location: → Supine → Back, Prone → Front [Position]
 - i Dependent (+)
 - ii Pressure = (-)
- e) Venous blood Pooling → distended → accumulated → deoxy Hb → blue - purple = Skin
- f) + shortly after death = 2 hr = Visible (Time Since Death)
 - i Max → 6-12 hrs
 - ii Fix → 7-8 hrs
 - iii Persist → Decomposition → Skin = Green color
- g) Moving body = absent
- h) Hanging body = glove stock appearance = in distal parts of hand and legs
- i) H_2S → bluish green

- j) CO → cherry red
- k) CN → bright red (Brick)
- l) Opium → black
- m) Nitro, Aniline and phosphorous → Chocolate - Brown

P A N ↓

4. Muscle stiffening

- a) Heat stiffening: $>65^{\circ}\text{C}$ = Coagulation of muscle - Protein
- b) Cold stiffening: $<-5^{\circ}\text{C}$ = Frozen body fluid
- c) Gas stiffening: Decomposition → GAS = 2-3 days
- d) Rigor mortis: ATP ↓
- e) Cadaveric spasm: ATP ↓



Rigor mortis = stiffening, Cadaveric-Stiffening

- a) Due to ATP ↓ Somatic death → cellular Death
- b) Takes some time → 1^o Relaxation → RM + → 2^o Relaxation
 - ↓
 - c) All muscles → Start → 1-2 hrs → Decomposition
 - i) Involuntary → 1st → Myocardium → Disappears first
 - ii) Voluntary
 - ↓
 - External
 - ↓
 - a) 1st → upper eye lid
 - b) Last feet [Toes] ← Last
 - c) Follows nysten law [Head to toe]
- iii) Rule of 12
 - 1. 12hrs = Appears in all → (M)

2. 12hrs = Remain in all → (M)
3. 12hrs = Disappears from All (M)
4. 36-48 hrs, Time since death

iv) Fetal < 7months = (-)

Onset Duration



- d) Violent death → early & short
- e) Strychnine death → early & long
- f) Arsenic death → late & long

6. Cadaveric spasm

- a) Aka Cataleptic rigidity
- b) Due to ↓ ATP
- c) Few deaths →
 - d) → Drowning
 - e) → Exhaustion (Antemortem Event)
 - f) → Firearm, fear
- d) Few m/s = voluntary = Hand
- e) Immediately after death = Instantaneous - Rigor
- f) 1^o relaxation = absent
- g) Not broken
- h) Manner of death

Somatic - Death

↓ (M)

Electric Stimuli



Response (+)

Late → Putrefaction (Decomposition)

1. Internal
2. Surest Sign of death
3. 1st → Brown → Intima → Aorta
4. 1st organ → Larynx - Trachea

Then

Liver - appearance (Foamy, Honey - Comb), lung

Brain → heart → kidney → uterus/prostate → FM/M



Bone (Teeth)



Non-gravid

5. Taylor → casper-dictum



a) Rate of decomposition



Air [1 wks] > Water [2 wks] > Soil [8 wks]

Decomposition

1. External
2. Surest Sign of death
3. Microorganism → Cl. Welchii
4. Enzyme/toxin → Leci thinase = Lytic

1st → Green color = location Right illiac Fossa



Sulf meth -Hb



Cecam

Summer → 12-18 hrs

2nd = Marbelling → superficial vein distended



Start +24hrs → thigh, chest, shoulder

Max 36-48 hrs

Color change: Green-brown due to → Sulf-Hb

3rd = maggot = Larva of Fly

- a) Time: 1-2 D
- b) Rainy season time: Early
- c) Study = entomology
- 4th =
- a) 3-4d

i. Face = Distend

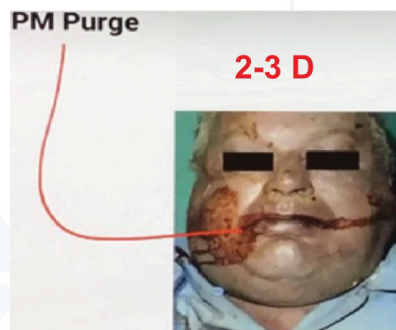
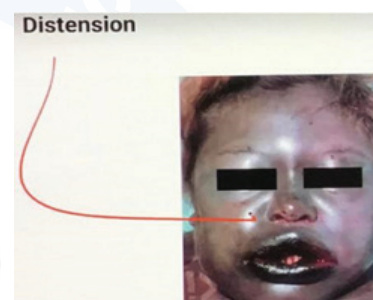
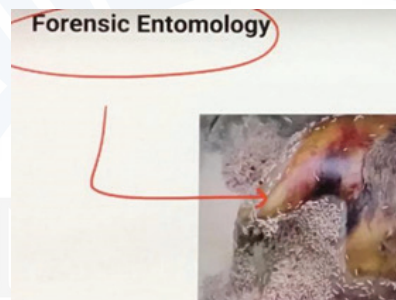
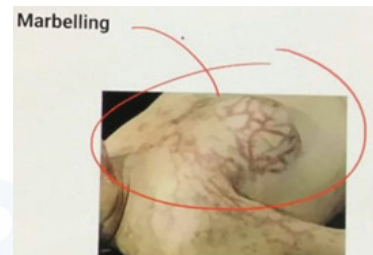
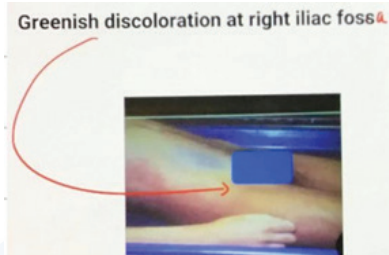
ii. Hair loose = 72-hrs

MIST

iii. Teeth loose = 72-hrs

iv. PM-purge = Vomit & Froth → Nostril / Mouth → 2-3d

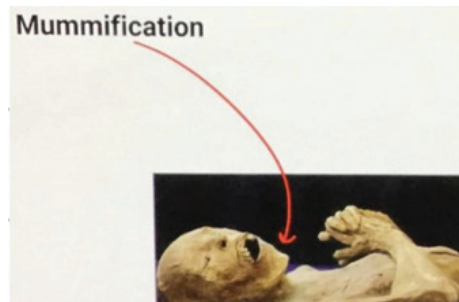
b) 5-10d = (Colliquative - necrosis)



Types:

1. Mummification aka Dessication

- Climate: Dry - Hot
- Water : Out → Shrink
- Smell: Odorless
- Natural features = Preserved
- Climate
- Time since death [3m - (1-2y)]
- Identification = easy



2. Adipocere

- ↓
- a) Aka Grave wax, Saponification
- b) Factors: Warm-humid/water/*Cl.welchii*/lipase
- c) Fatty part with above factors → adipocere
 - i. Fresh adipocere
 - 1. Greasy
 - 2. Trauslucent
 - 3. Ammonical smell (Rancid-Butter)
 - ii. Old adipocere
 - 1. Hard
 - 2. Dry
 - 3. Sweet smell
 - iii. Both are pseudosmell → Clothes
- d) 3d → 3wks [3m]
- e) Fetus < 7m → -ve
 - i. Climate
 - ii. Time
 - iii. Identification = easy (Lipase)



Starvation

a) M.char PM finding → Gall-bladder - Distension

Embalming fluid → body → to preserve → DB

Embalming fluid:

- i. Formaldehyde
- ii. Methanol
- iii. Phenol
- iv. Glycerine
- v. NaCl
- vi. Na citrate
- vii. Water

1. Ethanol = Not - Present
2. Femoral Vessels
3. Within 6hrs → Death
4. PM → Embalming

Unclaimed body → minimum preservation (72hrs) → _____

Medicolegal document → minimum preservation → 10_yrs

Declaration Definition



Geneva



Oath

Turkey

Tokyo

Torture → pt. = Istanbul Protocol [To-To]

Sydney

Death

Munich

Medicine

Helsinki

Human experiment

Oslo	Therapeutic abortion [0-0]
Venice	Terminal illness
Washington	Biological warfare
Lisbon	Rights of Patient



Active space



TOXICOLOGY

Chapter 17 :-

TOXICOLOGY

Founder → Orfila

Poison

Father = Paracelsus

1. Universal Ad → (Antidote) Adsorb
 - a) 2 Activated charcoal = Physical | Mechanical Ad, 1g/kg
 - b) 1 Tannic Acid
 - c) 1 Mg oxide
2. Coma - cocktail (opium / alcohol)
 - a) Ad:
 - i. Dextrose → Hypoglycemia
 - ii. Naloxone → O pium
 - iii B₁ = Thiamine → Ethanol
3. Hemodialysis = C.I:-
 - a) Kerosene
 - b) BZD → Ad → Flumazenil
 - c) Chloroquine, CuSO₄
 - d) Heroine → OP (Organophosphate)

Don = Digitalis

4. Gastric lavage

a) Ewald - Tube



Within → 3hr

Absolute

C.I:-

C → i. Corrossive = → Thin - Mucosa → Perforation

Carbolic acid = leathery mucosa G.L. → done

Relative C.I.

C → Convulsion

C → Coma

C → Hydrocarbon

↓
→ Kerosene

↓
Aspiration



Death = poison = PM

Toxicology analysis

Sample

Blood = 100 ml

Urine = 100 ml

Preservative

NaF, K⁺ Oxalate

Toluene, Thymol

Stomach + content	Rectified spirit → Best
Intestine = 30cm + content	MC → Common salt
Liver = 500gm + GB	
Half kidney - each	
Half spleen	

→ Histopath → Formalin

→ Virology → Glycerine

MIST

Arsenic → Cholera ← mimic (Acute)

Aldrich Mee's line → Nail

Mucosa: Red - Velvety Mucosa

Pigmentation: Rain - drop

Ad: Iron = Ferric Oxide

Chelating agent: BAL → Dimercaprol = British Anti Lewsite

Black foot disease

= F.D = 100-200mg

= Chronic → Bone, Skin, Nail, Hair



Mee's line



Thallium

Lead = Pb

Chronic poison = Plumbism, Saturnism

Findings:

Burtonian - line = Blue = Gums

Basophilic Stippling = → blood histopathology → Blue dot → RBC

MC = inhalation Child → Ingestion

Earliest, M.consistent → Face - Pallor → symptom, Constipation

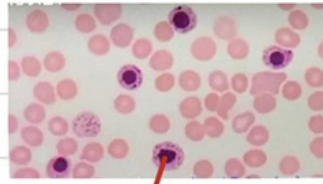
Chelating agent

1. Child = Encephalopathy → EDTA + BAL

2. Best → _____ = DMSA = Oral



burtonian line



Basophilic -
stippling

Phosphorus

1. White

- a) F.D = 60-120mg
- b) Smell: Garlicy, Crackers
- c) Aka Diwali-Poison Luminous → T/t Vit k
- d) Asterxis = Tremor → A
- e) Ad: CuSO₄ = Poison → C
- f) Phossyjaw

2. Red = (White + Red → Yellow)

- a) Non-toxic
- b) Smell: No
- c) Luminous: Non

Copper

- 1. Feature: Clapton Line (Green color in Gums)
- 2. Chelating agent : Peni - Cillanmine
- 3. Ad: K⁺ Ferro - Cyanide
- 4. Vineyard Sprayer lung disease

Cadmium → Bone → Itai → Itai Disease (Ouch - Ouch D.)

Iron → Desferi oxamine

Mercury = Hg

Japan → Minamata - disease

Eye change → Mercurialentis, Ant. lens Capsule

Erethism = SHY, Depressed [Psychiatric]

Chronic Hg → Hydra - Gyrism

Shake: Hatters - Shake

Glass - blower Shake

Tremor: Danberry

Acrodynia: Pink disease → Painful - Limbs

Ch.agent: BAL

Ad: Na-Formaldehyde Sulfoxylate

Ideal suicidal agent: CN

MC suicidal : Insecticide

Organophosphate

Smell: Kerosene

Inhibit → acetyl cholinesterase

Cholinergic

IOC → RBC-Cholinesterase

Ad: Atropine

Oxime (PAM)

Alkyl: HETP

OMPA

TEPP

Mallika → Malathion

Aryl : Parathion

Diazinon

Celphos = ALP3 → Phosphene

Inhibits → Cytochrome oxidase (IV)

Smell: garlicky smell

F.D = 1-3 tablet

No specific Ad

Asphyxiant gas

1. $\text{CO} + \text{Hb} \rightarrow \text{Anemic - Hypoxia}$

a) T/t

i. = O_2

b) PM stain: cherry red

2. H_2S [Sewer - gas]

a) Smell: Rotten - egg

b) Japan: Detergent - Suicide

c) Pm stain: blue green

3. CN^- [Prussic - acid], Apple-Seed

a) Smell: Bitter - almond

b) Inhibits cytochrome oxidase Histotoxic - Anoxia

c) Pm stain: brick red

d) Ad = Vit B12 $\rightarrow$ Best

Lilly Ad = Triple Ad

i. Amyl Nitrite

ii. Na Nitrite

iii. Na thiosulphate

e) Lee Jones Test

CNS depressants

Ethanol

1. $M = FM \uparrow = \downarrow \text{metabolizing enzyme} = \text{Alcohol} - \text{DH} \downarrow$

2. 90% - Denaturated Spirit

3. 95% - Rectified Spirit

4. 99.95% - Absolute Alcohol

5. Driving limit = 30mg - % $\rightarrow$ MACT 185

6. Nystagmus = 80mg %

7. Motor-incoordination = 150mg%

MIST

8. Coma-death = 300-400% Coma → Mcewan Sign → Pinch → Pupil → Dilated

9. Best test = GAS-Chromatography

10. Widmark - formula

Methanol → Illicit liquor → Hooch - Tragedy

1. Fatal dose = 80-120ml

2. Retina → Snow - field vision → Blind

3. TOC → Hemodialysis

4. Ad fomipazole (-) Alcohol - DH

Ethanol Comp. (-) Methanol

Ideal homicidal = P = Thallium → Mees-line → Alu - Puri → Sabzi



Garlicy

Alopecia, Polyneuropathy, Skin-rash



MC homicidal = P = Arsenic, Aconite

MC method of suicide = India → Hanging

MC method of para-suicide = Drug/Poison (attempted suicide)

Corrosive

Acid → Coagulative necrosis = Perforation

Alkali → Liqui factive necrosis (most dangerous) Spasm

1. Boric acid causes → Boiled lobster Synd.

2. Nitric acid = Aqua-Fortis

a) Tissue → Yellow color due to Xanthoproteic. Rxn

3. Oxalic acid

a) Use → Ink - Remover

b) Component: Ca^{++} Oxalate

c) Decreases Ca^{++} in Blood

- d) Ad: Ca^{++} gluconate, wall - Scrapping (Ca^{++})
- e) Coffee-ground - vomitus
- 4. Carbolic acid = phenol, Marasmus ← Mimic
 - a) Pupil = Cons.
 - b) Cartilage = Ochronosis
 - c) Urine = Carboluria → Green color urine
 - d) Phenol antiseptic → decomposition delayed
 - e) Mucosa appearance = Leathery
 - f) Gastric lavage can be done
- 5. Strongest = H_2SO_4 , Sulphuric - Acid
 - a) Oil of VITRIOL
 - b) F.D = 5-10ml
 - c) F.P = 15-20hr
 - d) Ad: Mgo
 - e) Appearance
 - i. Mucosa/organ: Black aka Blotting Paper mucosa
 - ii. Teeth : Chalky - White
 - f) MC = vitriolage = H_2SO_4 [Acid-attack)
 - g) MC plant vitriolage: Semicarpus

CNS stimulant = Marijuhana

1. Cannabis

- a) Component: 9 - Tetra - Hydro - Cannabinol (9-THC)
- b) Charas = M.potent = Resin
- c) Diazepam = Ad
- d) Excite = whole part of plant
- e) Female flower → Ganja, Joint, Reefer, Maal, Weed
- f) Run Amoke = homicide → suicide

MIST

- g) Bhang = dry leaves; smell: Burnt - Rope
- h) Hashish → CHARAS - Amotivation - Synd.
- i) Bhang → Sweet → Majoon

2. Strychnine

- a) aka Nux-Vomica, Kuchila, Dog-Button
- b) Affects : Spinal - Cord = A.H. Cells → Renshaw → Post-Synaptic → (-)Glycine
- c) F.D → 1-Crushed-Seed
- d) Causes R. M. Spasm → Clonus → Asphyxia
- e) MC position of body: Ophisthotonus Hyperext. of Back → Bow

Mimic - Tetanus

SPINAL CORD-STRYCHNINE



Concavo - Convex



Seeds

Cardiac poison

1. Aconite / Monk hood = Blue Rocket, Metha Zaher

- a) Hippus sign = Alternate → C/D → Pupil

=  Root M. Toxic

= K^+ ↑

COD = Car. Arrhythmia.

Monks hood



2. Yellow oleander → Peela - kaner

- a) = Cerebra - Thevetia

→ Cerebrin

→ Thevotoxin

Yellow Oleander



→ Bell Flower

3. White oleander

a) Pink →

= Kaner →



b) (Nerium - Odorum)

→ Nerine

→ Oleandrin

4. Nicotina tabacum = Tobacco MC substance use / abuse in India

a) → illicit → Cannabis

Datura → Blind - bat, hot - hare, dry - bone, Red - beet, Mad-hen

Delirium

9D - Poison

Anticholinergic:

→ Hyoscine

MC known as= Robbery / Railway, Road poison

FD = Hundred - Seed

Mydriatic = Cornpicker pupil

D.Alba = White color

D.Niger = Black color

D.Stramonium = Thorn apple, Jimson-weed, devils trumpet

Ad:

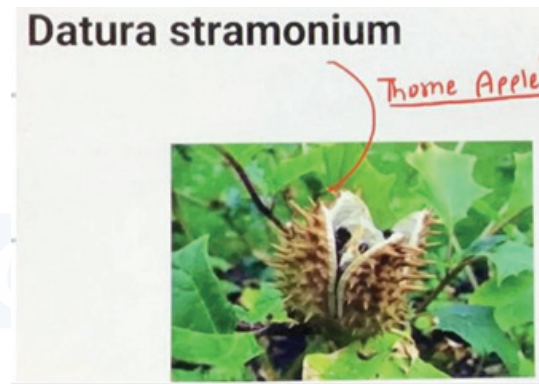
Imaginatory Threads



Carphologia

1. Physotigmine

2. _____



Atropa Bella dona → Deadly night-shade

Opium → Somniferous

Aka Afim = Kasoomba

Poppy fruit → Un-Ripe fruit = Milk = Latex → Brown = Opium

Pupil : Coma → Pin-Point

Ad: Naloxone

Seed = Khas-Khas

Diacetyl morphine → Semisynthetic

Heroin, junk, dope, brown sugar, smack

Heroin + cocaine = Speed - Ball [He + She]

Heroin + strychnine = Hot - Shot

Heroin smoking = Chasing the Dragon

→ Crack

Cocaine → tongue/teeth color: Black

→ Coke

Cocaine causes Nasal - Septal perforation

→ She

Cocaine → sample = Nasal-Swab

→ White-Lady

Ad: Amyl - Nitrite

→ Snow

Magnan symptoms: Tactile Hallⁿ → Cocaine-Bug

Sympathomimetic → BP, HR, RR ↑

Red - color → berries



Plant poison = Irritant

1. Ricinus

- a) Castor, Arandi
- b) FD = 10-Seeds
- c) Toxin = Ricin

2. Croton

- a) Napala, Jamalgota
- b) FD = 4-5 → Seeds

3. Abrus, Ratti, rosary beads = Gunja

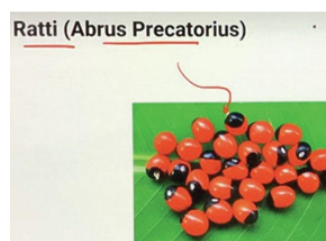
- a) FD = 1 - Seed
- b) Abrin = Viper-Venom
- c) Animal poison = kill = sui = needle

Ideal - Cattle - Poison

- d) Arrow poison
- e) Abortion poison
- f) T/t → dissection of sui

i. Anti Abrin

ii. Gastric lavage



4. Ergot = claviceps, Fungus = Rye

a) Hallucination: Tactile = Formication



Snake poison

Ophiotoxemia

1. Neurotoxic = Elapidae

Hood +

a) Cobra = NAJA - NAJA

c). King-Cobra = *Ophiophagus*

i. Spectacle → Hood

→ Fatal → Most

ii. (Breath Count Test) → Neurotoxic

iii. Respiratory m/s paralysis → COD

b) Common krait *Bengarus*

i. White Scale

ii. Respiratory m/s paralysis → COD

Krait [Minimum Local Symptom] Tt = Mechanical Ventilation + ASV

Cobra = MV+ASV+Neostigmine + Atropine

2. Vasculotoxic[hemolysis, hemorrhage], Bleeding → DIC

a) Viper

Tt = ASV + FFP + SX - Debridement

Max - Local Symptom → Pain Swelling, bleeding, Gangrene, Necrosis

"Whole Blood clot test" → Viper

i. Russel (triangle head) = Brown - Scale

ii. Saw-scale = Coiled

Tt →

- a) Anti-snake venom = Cobra + Krait + R.V. + S.S.V.
- b) Bombay = 8-10 Vials, I. V.
- c) Hoffkin institute

Reassure

Immobilize

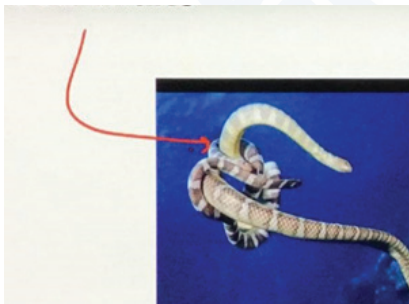
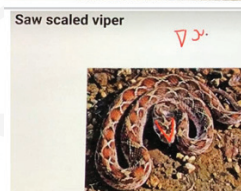
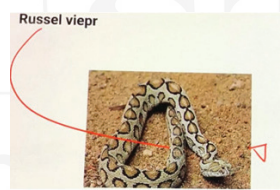
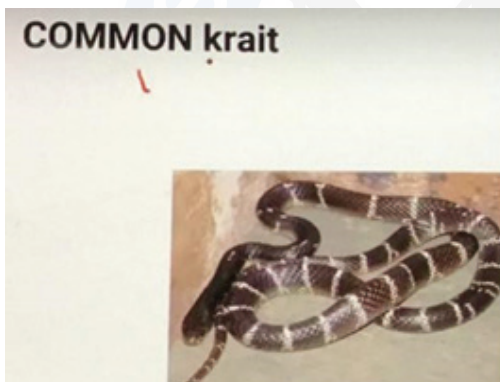
Rx → Go

Hospital, History, Symptom

Tt,

3. Sea snake → Myotoxic → M. Necrosis → Renal-Failure

a) Muscle-paralysis (Skeletal muscle)



Active space

Blister Beetle

= Spanish Fly

Bites releases Cantharide poison → Priapism → painful erection of penis

→ Arachnida

→ 8 Legs

→ Neuro Toxin → Scorpion, Spider

→ Priapism

Scorpion bite = Autonomic - Storm = S ↑ + PS ↑

→ Prazosin, ASV

Spider bite = Dapsone + ASV

M.poisonous spider = Black - Widow

Calotropis = toxin = Calotoxin & Uscharin

Madar, AKDO, Rubber - Bush

Semicarpus = Bhilawa

Marking nut

1. Semicarpol

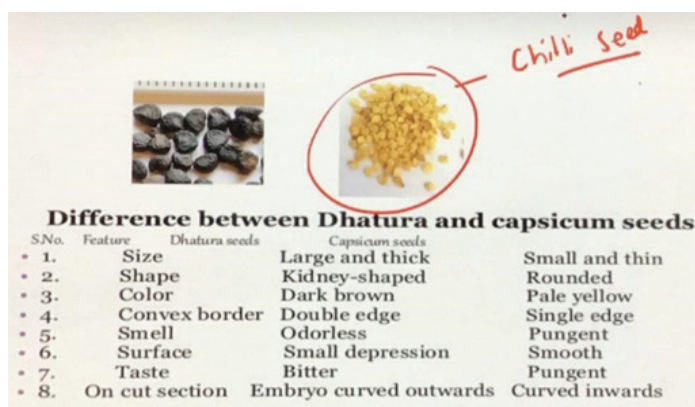
2. Bhilawanol

+ heart = seed



Chilli seed = Capsicum

= Hunan - Hand = Dermatitis



Nasal inhalation

1. Sniffing: Container = inhale
2. **Huffing**: inhalant-**cloth** = inhale
3. Bagging: plastic bag → inhale
4. Ballooning : balloon = gas → inhale
5. **Glading** = **spray room** → inhale
6. Dusting = inhale aerosol → directly

Lie detection technique

1. Polygraph test → BP, Pulse, RR, HR, Skin/m.
2. Truth serum drug → Thiopentone > Hyoscine
3. Brain mapping [best]

Body packer syndrome = Mule Synd.

1. Drug - Smuggling → Large → Condom, Polythene
2. _____ ↓
3. Risk ↓ [Planned] Swallow, Cavity

Body stuffer syndrome

1. Drug → Conceal

Risk ↑

Acid

Poison	Smell	Pcm	LSD	Dry-wine
Nike, Nitrobenzene →	Shoe-Polish	Ad	Serotonin ↑	Chloral-Hydrate
Para-Aldehyde	Acrid	N.Acetyl	Trip=Good, Bad	Micky-Finn,
Conium →	Mousy	Cysteine	Flash Back PHn	Knock-out Drops
Nerve-Poison			Hallu ⁿ → Erotic	

Ethanol = 15y= Quit → 2-3D

→ Delirium Tremens = Crime = NR

→ De = Disorientation = Time, Place, Person

Anxious ← A = Autonomic Act ↑ = BP ↑ HR ↑ RR

Tremor = Coarse

Hallucination = Visual

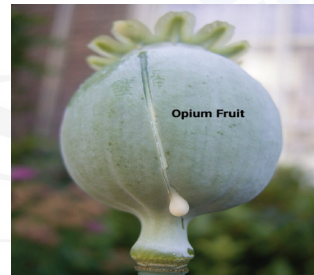
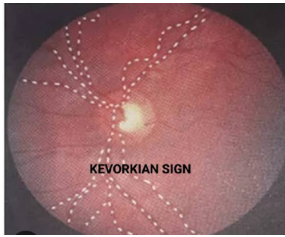
Ethanol = Vit -B₁ ↓ - Global Confusion

Wernicke Encephalopathy - Ophthalmoplegia

G O A AtaxiaB₁ ↓ Korsakoff Psychosis → Confabulation → False - StoryCAP

→ Amnesia

→ Polyneuropathy



Active space

Workbook Question**ASPHYXIA**

1. Gettler's test is done for death for?
 - A. Drowning
 - B. Hanging
 - C. Burns
 - D. Phosphorus poisoning
2. Best indicator of ante mortem drowning is?
 - A. Froth in mouth and nostrils
 - B. Cutis anserina
 - C. Washerwoman's hand
 - D. Water in nose
3. All are traumatic asphyxia except
 - A. Railway accident
 - B. Road traffic accident
 - C. Accidental Strangulation
 - D. Stampede
4. Sexual asphyxia is seen in?
 - A. Transvestism
 - B. Frotteurism
 - C. Masochism
 - D. Tribadism
5. In asphyxia organ to be dissected last-
 - A. Heart
 - B. Neck
 - C. Abdomen
 - D. Thorax

FORENSIC PSYCHIATRY

1. Testamentary capacity refers to ?

- A. Ability to make a valid will
- B. Criminal liability
- C. Right to vote
- D. Ability to give evidence

2. In which of the following conditions is a person held responsible for a crime?

- A. Alcoholic paranoia
- B. Delirium tremens
- C. Voluntary drunkenness
- D. Dhatura intoxication

3. All are features of 'feigned Fake insanity', EXCEPT:

- A. Sudden onset
- B. Predisposing factors present
- C. Motive present
- D. Insomnia cannot persist

IDENTIFICATION

1. Palatal print commonly taken from

- A. Anterior part of palate
- B. Posterior part of palate
- C. Lateral part of palate
- D. Middle part of palate

2. The most reliable criteria in Gustafson's method of identification is

- A. Root resorption
- B. Attrition
- C. Cementum apposition
- D. Transparency of root

3. Stack Formula is used to estimate age through dentition is for.

- A. Infants
- B. adults 25-50yrs
- C. adult >50yr
- D. elderly

4. 1st ossification centre appears at:

- A. Beginning 2nd I/U month
- B. Beginning of 3rd I/U month
- C. End of 3rd I/U month
- D. 4th month of gestation

5. Corporo-basal index is used to determine

- A. Race
- B. Sex
- C. Age
- D. Stature

INJURY

1. Criteria for grievous hurt are all EXCEPT

- A. Endanger to life
- B. Privation to eye/sight
- C. Permanent disfiguration of face
- D. Injured person admitted in hospital for 21 days

2. Beveling of skull is seen in

- A. Drowning
- B. Hanging
- C. Fire arm injury
- D. Road traffic accident

3. Rupture of the skin and subcutaneous tissue is called

- A. Incised wound
- B. Abrasions
- C. Lacerated wound
- D. Contusion

4. Brush burn is caused by

- A. Abrasion
- B. Contusion
- C. Laceration
- D. Firearm injury

5. In whiplash injury, damage is caused to

- A. Skull
- B. Spine
- C. Rib cage
- D. Long bones

LEGAL PROCEDURE

1. Not a grievous injury:

- A. Multiple scars of face
- B. Fracture of femur
- C. Emasculation
- D. Contusion of breast

SEXUAL OFFENSE

1. Which of the following is not punishable if done with consent of women:

- A. Sodomy
- B. Adultery
- C. Incest
- D. Sexual intercourse below the age of 18 year

2. Bestiality is a

- A. Beast like behaviour of an insane
- B. Cruelty to animals
- C. Cruelty to fellow humans
- D. Having intercourse with animals

3. Statutory rape is rape on girl below;

- A. 10yrs
- B. 12yrs
- C. 15yrs
- D. 18 yrs

4. Incessant sexual desire in males:

- A. Satyriasis
- B. Nymphomania
- C. Vaginismus
- D. Quoad hoc

5. According to the 2013 amendment the age for sexual consent is

- A. 16 years
- B. 18 years
- C. 20 years
- D. 15 years

ASPHYXIAL DEATH

1. Most common site of knot in case of judicial hanging is
 - A. Lateral
 - B. Below chin
 - C. Occiput
 - D. Below ear.
2. On examination a dead guy was found with a ligature mark on neck. According to police, the person's legs were touching the ground while hanging. You suspected death due to partial hanging. This death is due to weight of around
 - A. 1.15 kg
 - B. 2.20 kg
 - C. 3.10 kg
 - D. 4.50 kg
3. Hanging with the feet touching the ground is seen in:
 - A. Partial Hanging
 - B. Complete Hanging
 - C. Homicidal Hanging
 - D. Suicidal Hanging
4. Cause of death in hanging is-
 - A. Venous congestion
 - B. Asphyxia
 - C. Cerebral anemia
 - D. All of the above

5. In judicial hanging, the knot comes to rest at the:

- A. Occiput
- B. Chin
- C. Angle of mandible
- D. Below the cheek

THERMAL INJURY

1. Which of the following tissues is most resistant to electric current?

- A. Muscle
- B. Skin
- C. bone
- D. blood

2. Which of the following is the most characteristic feature of a joule burn?

- A. It is round in shape
- B. It has a depressed base
- C. there is an area of blanched skin at the periphery
- D. It is usually green in color

3. Vesicles in an ante-mortem burn are differentiated from a post-mortem burn by the presence of

- A. Chloride
- B. Albuminous fluid and chloride
- C. Plasma and wbc in fluid
- D. Air

TOXICOLOGY

1. Which heavy metal causes constipation
 - A. Lead
 - B. Mercury
 - C. Copper
 - D. Arsenic
2. In case of burning fire in a closed room, death occurs due to
 - A. CO₂ poisoning
 - B. CO poisoning
 - C. Cardiac failure
 - D. Respiratory failure
3. Portrait Parle means
 - A. Bertillionsyotm
 - B. Galton syotem
 - C. Decomposition Rate
 - D. Casper dictum
4. Rectified spirit (preservative) contains
 - A. 99.9% alcohol
 - B. 95% alcohol
 - C. 90% alcohol
 - D. 50% alcohol
5. DOC for organophosphorus poisoning is
 - A. EDTA
 - B. BAL
 - C. PAM
 - D. All of the above

Answers

ASPHYXIA: Ans : 1-A, 2-A, 3-C, 4-C, 5-B

FORENSIC PSYCHIATRY: Ans : 1-A, 2-C, 3-B

IDENTIFICATION: Ans : 1-A, 2-D, 3-A, 4-A, 5-B

INJURY: Ans : 1-D, 2-C, 3-C, 4-A, 5-B

LEGAL PROCEDURE: Ans : 1-D

SEXUAL OFFENSE: Ans : 1-C, 2-D, 3-D, 4-A, 5-B

ASPHYXIAL DEATH: Ans : 1-B, 2-D, 3-A, 4-D, 5-C

THERMAL INJURY: Ans : 1-C, 2-C, 3-B

TOXICOLOGY: Ans : 1-A, 2-B, 3-A, 4-B, 5-C



INDIAN LAWS CHANGED

- IPC changed to BNS (Bhartiya Nyaya Sanhita). A total of 358 sections
- Cr Pc changed to BNSS (Bhartiya Nagarik Suraksha Sanhita). A total of 531 sections
- IEA changed to BSA (Bhartiya Sakshya Act)

Important BNS = Definition, Punishment

Category	BNS
1. Various Definition	2 BNS
2. Criminal Responsibility (based on age, mental illness and intoxication)	20 -24 BNS
3. Consent	25 -30 BNS
4. Sexual offences	63 -79 BNS
5. Offences Related to marriage	80 -87 BNS
6. Criminal Abortion	88-92 BNS
7. Offences against child	93 -99 BNS
8. Offences affecting life (Murder, Suicide etc)	100 -113 BNS
9. Hurt, grievous hurt etc.	114 -125 BNS
10. Criminal force, assault etc	128 -136 BNS
11. False evidence	227 -246 BNS

2 BNS: Various Definitions

- Subsection 6 of 2 BNS: Death
- Subsection 14 of 2 BNS: Injury
- Subsection 17 of 2 BNS: Life

20 -24 BNS: Criminal Responsibility (based on age, mental illness and intoxication) **RESPONSIBILITY-AT-20**

- 20 BNS: Child < 7 years, not responsible (Minimum age for C.R)
- 21 BNS: 7-12 years, +/-responsibility, court decision
- 22 BNS: Criminal responsibility of Mentally ill (McNaughten's)
- 23 BNS: Involuntary intoxication
- 24 BNS: Voluntary intoxication

25-30 BNS: Consent(7)

- 25 BNS: Act not intended and not known to likely to cause death or grievous hurt, done by consent. **Age >18Y, Surgery**
- 26 BNS: Act not intended to cause death, done in good faith for person's benefit by consent (e.g. surgeon doing risky surgery with consent).
- 27 BNS: Act in good faith for a person <12 years of age, or mentally ill by consent of the guardian.
- 28 BNS: Consent under fear, misconception, < 12 year, mentally unsound, intoxication not valid
- 29 BNS: Consent for acts which are offence is not valid (e.g. criminal abortion)
- 30 BNS: Act done in good faith for benefit of a person "without consent" in Emergency

63 -79 BNS: Sexual Offences

- 63BNS: Def. of Rape
 - Minimum age of girl for consent for sex: 18 yrs
- 64 BNS: Punishment for rape (10 years to Life) (6+4=10)
- 65BNS: Punishment for Rape in certain cases
 - < 16 yrs of girl (20 years to Life)
 - < 12 yrs of girl (20 years to Death)
- 66BNS: Causing death/ vegetative state of victim while committing rape (20 yrs-Death)
- 67BNS: Sex by husband during separation (2-7 yrs)
- 68 BNS: Sex by person in authority (5-10 yrs)
- 69 BNS: Sex by using deceitful means, e.g. false promise (6-9) of marriage (upto 10 years)

70 BNS: Gang rape (6-7)

- > 18 years (20 yrs -life)
- < 18 years (Life imprisonment-death)
- 71BNS: Repeat offender (Life imprisonment -death)
- 72BNS: Disclosure of identity of rape victim (upto 2 years)
- 73 BNS: Publishing any matter related to court proceedings (upto 2 years)
- 74BNS: Sexual assault by criminal force (1-5 yrs)
- 75BNS: Sexual harassment
 - a) Physical contact / request for sexual favours/ showing pornography (upto 3 years)
 - b) Sexually coloured remarks (upto 1 year)
- 76 BNS: Removal of clothes of woman (3-7 years)-DISROBING
- 77 BNS: Voyeurism → See - See (77)

- o 1st Offence (1-3 years)
- o 2nd Offence (3-7 years)
- 78BNS: Stalking
 - o 1st Offence (upto3 years)
 - o 2nd Offence (upto5 years)
- 79 BNS: Word, gesture or act intended to insult modesty of woman. (upto3 years)

80 -87 BNS: Offences related to marriage

- 80BNS: Dowry Death. It is death of women within 7 years of marriage (7 yrsto life) $4+4 = 8$
- 85 BNS: Women subjected to crueltyby husband or in laws (upto3 years)
- 88-92 BNS: Criminal Abortion (8+8)
- 88 BNS: Criminal abortion with mother's consent
- 89 BNS Abortion without mother's consent
- 90 BNS: Deathof mother due to abortion
- 91 BNS: Act done with intent to prevent child being born alive or to cause to die after birth.
- 92 BNS: Causing death of quick unborn child by act amounting to culpable homicide eg.Hitting abdomen

93 -99 BNS: Offences against children

- 93BNS: Child <12 yrsabandoned by parents ($9+3+=12$)
- 94BNS: Newborn death concealment

100 -113 BNS: Offences affecting Life (murder, suicide etc.)

- 100 BNS: Def. Of Culpable homicide - 100
- 101 BNS: Def. Of Murder

- 103 BNS: Punishment for Murder
- 105 BNS: Punishment for Culpable homicide not amounting to murder (Provocation → Paanch)
- 106 BNS: Death due to negligence (Punishment up to 5 years; but for medical negligence up to 2 years)One-g
- 107, 108 BNS: Abetment of suicide
- 109 BNS: Attempt to murder
- 110 BNS: Attempt to culpable homicide

114-125 BNS: Hurt, Grievous Hurt

- 114BNS: Def. Of hurt
- 115 BNS: "Voluntarily" causing hurt (1 year)/10K → P
- 116BNS: Def. Of grievous hurt
- 117 BNS: "Voluntarily" causing grievous hurt7Y → P
- 118 BNS: Hurt or G. Hurt by "dangerous weapon"(3Y 10Y) → P
- 122 BNS: Hurt or G. Hurt on "Provocation" (PANCH)
- 123 BNS: Hurt by "Poison" (6)
- 124BNS: G. Hurt by Acid (Throwing or administrating etc.)
FENKA (One two ka Four)
- ATTACK → 10y-L.i.
- **5y-7y-ATTEMPT VITRIOLAGE**

128 -136 BNS: Criminal Force and Assault

- 129: Defines criminal force
- 130: Defines Assault (attempt or threat to use criminal force)

227 -246 BNS: False Evidence

- 227BNS: Perjury definition (Giving false evidence)
- 228 BNS: Fabricating false evidence (F+F=12)
- 229 BNS: Punishment for perjury (false evidence-upto 7 years)
- 234BNS: Issuing False certificates 2-3-4
- 238 BNS: **disappearance** of evidence of offence (13)
- 296 BNS: Obscene acts (Exhibitionism) - DO NE SIX

Important BNSS Procedure, Investigation

- **2 BNSS:** Various definitions (including Definitions of various types of offences, like bailable, cognizable, warrant case, summon case etc) offences
- 33 BNSS: Crime information to police
- 51 BNSS: Med Exam of arrested person for any offence (forceful examination allowed)-**PRISONER** → Paanch
- 52 BNSS: Med Exam of arrested person accused of rape (forceful examination allowed)
- **63 -71 BNSS:** Summons
- 184 BNSS: Medical examination of a victim of rape
- 194BNSS: Police Inquest-4
- 196BNSS: Magistrate inquest-6

Important BSA = Witness, evidence

- 26: Dying Declaration (2+6)
- 39 → Expert witness
- 110 → Presumption of Survivorship
- 111 → Presumption of Death (NELSON)



7 Yr. (Not Seen)